A THESIS ON

VOLLEY

Design and Computational Feasibility Assessment of a Commanded Electromagnetic CubeSat Deployment Mechanism

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SUBMITTED BY

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Abstract

This final-year project report evaluates a magazine-fed electromagnetic CubeSat deployer as a *computational design study*. For a twelve-payload 3U configuration, the historical periodic-force model predicts a 16.029 m/s release at 10.068 g and 2.782 kJ gross shot draw. A finite-array/stator 3-D analytic integral challenges that speed prediction, giving 12.448 m/s under ideal phase. A separate 2-D finite-element screen gives 1.082 kJ ideal work; an independent numerical 3-D surface-charge formulation reproduces 1.042 kJ under shared assumptions. An illustrative 14 mm face gap lowers ideal work to 0.907 kJ, while a conditional full-winding bank rerun gives 2.099 kJ gross draw. These are model results, not a selected motor rating. An independent Cartesian two-body calculation reproduces a 28.800775 km immediate semi-major-axis rise for the imposed historical input at 450 km. A native FreeCAD 1.0 review document and STEP exports expose an 11 mm transverse shortfall in the specified side-fed layout, with 32,915 mm^3 of exact-solid overlap per cassette. The estimated 126.6 kg dry mass gives 10.547 kg of deployer per carried 3U, failing both the 2 kg economic screen and an approximately 6 kg spring-canister parity comparison. Atmospheric lifetime, exact payload interfaces and full mission delivery remain conditional or open. There is no physical prototype or VOLLEY test data. The Gen5 result is a documented model-based decision against this 3U configuration. A separately versioned Gen6 design study selects a guided lead-screw feeder with fixed dedicated cells, capped-channel containment and one indexed receiver; its analytical and nominal-geometric evidence is distinguished from physical qualification. High-speed scaling remains future research.

Keywords: CubeSat deployment; electromagnetic launcher; systems engineering; numerical verification; CAD interference; orbital mechanics; model credibility.

Abbreviations and evidence convention

Table 1: Abbreviations and evidence convention — Term.

Term	Meaning
CAD / STEP	Computer-aided design / ISO 10303 solid-geometry exchange
CG / ICD	Centre of gravity / interface control document
FEM / FEA	Finite-element method / finite-element analysis
J2 / LEO	Second zonal gravity term / low Earth orbit
LSM / OTV	Linear synchronous motor / orbital transfer vehicle
1U / 3U	CubeSat form-factor classes; no specific spacecraft is qualified

All plotted VOLLEY results are computations unless explicitly described as external experimental data. CAD views and Blender renders show geometry rather than manufactured hardware. “Failed” identifies a missed evaluated criterion; “unverified” identifies behavior unresolved by the stated method. Numerical verification is distinguished from physical validation throughout.

This issue contains the evaluated Gen5 study and separately identified Gen6 research and the FM08-R3 feeder design selection through 23 October 2026. It is issued for first similarity review; institutional approval and the similarity result are not asserted. Gen6 research does not alter the evaluated Gen5 configuration.

CHAPTER I

Introduction and problem definition

1.1 The deployment problem

The primary mission determines the launch vehicle's delivered orbit. Secondary CubeSats may have limited onboard propulsion, mass, power, or programmatic freedom to correct that orbit. Standard canisterised spring systems are mature and effective at clearance and separation. Their release conditions, however, are governed by spring energy, payload mass, and the device rather than a separately commanded velocity profile for every satellite.

The distinction between **release timing** and **relative-state change** is central. Waiting alone between zero-relative-impulse releases from an unchanged host does not produce persistent in-track phase spacing. Spring impulse, differential drag or a host maneuver can produce a relative state; a commanded velocity along or against orbital motion can change the satellite's orbital energy and semi-major axis. VOLLEY studies the latter capability while retaining a clear comparison with spring release and other alternatives. It does not claim that all constellation phasing requires a motor.

The host is an active participant in the mission model. After its primary mission, a suitable hosted platform would need attitude control, available resources and permission to operate. It could perform coarse orbital repositioning; VOLLEY would assign each secondary payload a local release state. No specific upper stage has been selected or approved for that role.

1.2 Buyer and spacecraft fit

The audited market and customer-fit report checks the relevant Drive market thesis and independent business-case review against current primary sources. Planet's FY2026 filing and Spire's 2025 filing show repeatable, configurable smallsat production. NASA's 2026 launch review confirms that a primary mission can constrain the orbit available to rideshare payloads. Together, these support a *mission hypothesis*, not customer demand.

The likely party that could pay for individualized release states is a carrier, rideshare integrator or constellation operator responsible for a delivered manifest. A standardized satellite-bus maker may specify interfaces but need not buy the system. Replenishment batches and mixed-payload rideshare are candidate jobs. A one-satellite mission already in a suitable orbit is a poor fit and should favor a qualified conventional dispenser. The analyzed Gen5 system carries twelve 3U spacecraft but fails its own 3U mass screen; a PocketQube batch is a **future interface redesign**, not an accommodated or qualified Gen5 payload class. Payloads sensitive to magnetic fields need an explicit exclusion or design remedy. No buyer, payload or provider has confirmed these requirements.

An economic comparison requires the same manifest delivered with ordinary spring impulse, spring plus host manoeuvres, onboard propulsion, an orbital-transport service and VOLLEY. The published SpaceX rideshare starting price cannot stand in for an installed VOLLEY cost or savings quote. A three-to-five-year replacement cycle and a 5–50 m/s release range are research hypotheses from the Drive market report, not Gen5 operating requirements or established market demand. The Gen6 1 km/s upper objective is a different, unvalidated regime.

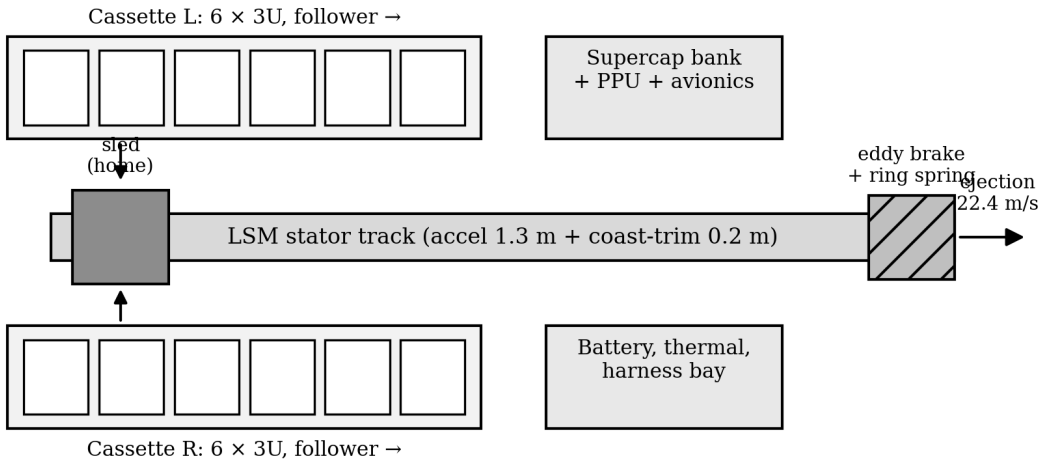


Figure 1: Gen5 layout and mission hardware elements. Diagram from the thesis analysis.

1.3 Research question

The analytical gap is not “no one has considered electromagnetic deployment.” It is the coupled trade between: (i) commanded release, (ii) payload-side interface and acceleration, (iii) a reusable feed/release architecture, (iv) host reaction and mission utility, and (v)

installed system mass and faults. A speed-only comparison can make a concept appear attractive while omitting its sled, power bank, brake, magazine, enclosure, control, and retained mass after each shot.

The thesis addresses that gap by constructing a documented Gen5 reference configuration and testing its own claims against explicit comparisons. It also records when those tests weaken the concept. Physical viability remains a separate research gap: numerical model agreement cannot establish manufactured tolerances, actual force, payload cleanliness, cycle reliability or provider compatibility.

CHAPTER II

Objectives and study scope

Table 2: Objectives and study scope — Analytical research objective.

Analytical research objective	Completed outcome	Interpretation
Model a commandable 3U system	Reference CAD, magazine concept, sled, motor, arrest and power architecture	Gen5 mechanics retain their evaluated limitations; the separate FM08-R3 feeder design is documented in Appendix C
Quantify the release	Historical periodic shot, circuit and closed-loop simulations; adverse finite-force screen	No selected or verified speed rating
Evaluate orbital value	Historical-input orbit model and selected numerical cross-checks	Conditional geometry; rated lifetime remains open
Test whole-system competitiveness	Mass rollup, spring/alternatives comparison and preset criteria	Gen5 fails the 3U mass target
Challenge model credibility	Numerical verification, defect corrections and provenance record	Scope and remaining physical uncertainties made explicit

The objectives define an evaluated computational study. Completion of an analysis is distinguished from satisfaction of a design requirement. A failed mass or fit criterion remains a result; it does not establish flight readiness. No physical prototype or experimental VOLLEY measurements are reported.

CHAPTER III

Literature review and research contribution

Table 3: Literature review and research contribution — Approach.

Approach	Established capability	Relevance and limit
Spring deployers	Flight heritage and simple separation near the low-m/s range [1]	Strong benchmark for reliability and installed mass; release energy is largely built into the device.
Spring release and differential drag	In-track spacing and gradual orbital adjustment [3]	A relative impulse or drag difference creates the separation; waiting alone from an unchanged co-orbital host does not.
Orbital transfer vehicles	Coarse orbital repositioning through a propulsive carrier	Valuable comparator for whole-mission delivery; greater spacecraft and integration burden must be counted.
Electromagnetic CubeSat studies	Magazine transport and electromagnetic launch have already been studied [4–6]	Some configurations impose conductive armatures, high acceleration or different maturity; novelty cannot be claimed for electromagnetic deployment alone.

A high-speed electromagnetic comparator summarized in the project literature record is Feng *et al.* [4], which models a 321.56 m/s launch of a 20 kg CubeSat but at acceleration on the order of thousands of g and with an armature/interface issue. Zhao *et al.* [5,6] study stacked CubeSat transfer and storage, including measured transport behavior in a related mechanism. These works establish that electromagnetic delivery and magazine transport

are prior art [4–6]. VOLLEY's more limited question is whether a **commandable** release can be delivered to an **unmodified** satellite at a much lower acceleration while accounting for the whole installed system. No universal CubeSat quasi-static acceleration qualification is assumed; the launch provider and payload-specific limits must govern a real design [2,8].

3.1 Closest feeder and deployer studies

The full-text feeder comparison examines the closest 2022 and 2025 studies with page references and publisher-source hashes. The 2022 electromagnetic release prototype reports gravity-compensated dummy releases, not VOLLEY performance. The 2025 storage prototype uses spacecraft-side grooves, guides and iron patches; this does not establish the unchanged passive-payload objective adopted here. Its 1200 N receiving-force result is simulated, while its fast laboratory pushes are gravity-assisted. The distinction requires actual payload interface approval, contact/load analysis and shared-path isolation in future VOLLEY work. Broader end-effect, mission and carrier literature and independent technical review remain outstanding.

The numerical results reported in [4] are contextual literature evidence and have not been independently reproduced here. The full-text inspection of [5,6] supports the detailed interface and test-condition comparison above. A shared release principle does not transfer another device's measured performance or reliability to VOLLEY. The defensible contribution is the coupled, reproducible feasibility assessment of the stated Gen5 configuration, including its adverse findings.

CHAPTER IV

Research methodology and governing models

The project follows an engineering evidence chain:

1. **Mission/use-case framing:** specify the payload, host responsibilities, target orbital change, safety limits and competent conventional comparators.
2. **Architecture trade:** examine force production, loading, restraint, guidance, arrest, reset and fault behavior, rather than only ideal acceleration.
3. **Geometry and installed mass:** build CAD geometry and count the sled, arrays, enclosure, feed, power, controls and other installed parts.
4. **Coupled models:** compute magnetic field and thrust, shot dynamics, power/circuit behavior, control dispersion, structural/flow responses and orbital consequences.
5. **Challenge the claims:** use alternate numerical methods and acceptance bands, preserve misses and update the baseline only with a recorded reason.
6. **Decision:** compare orbital benefit against mass, reliability, payload/host compatibility and the available alternatives.

4.1 Controlled question and acceptance logic

The evaluated configuration is the twelve-3U, side-fed Gen5 reference. Its mission and mechanical claims cannot be transferred to the later R1 enclosure or Gen6 first-article candidates without new analyses. The study asks whether this reference yields a credible commandable release and an installed advantage while fitting its stated geometry. A failed criterion is a completed result of that question; it is not a validated deployer.

Table 4: Controlled question and acceptance logic — Decision question.

Decision question	Test and evidence	Acceptance interpretation
Can the specified side-fed parts coexist?	Native FreeCAD/STEP exact-solid checks and width arithmetic (P116)	Nonintersection and positive service clearance required; reference **fails** before tolerance stack

Decision question	Test and evidence	Acceptance interpretation
Does the historical 3U speed survive finite force geometry?	Periodic model versus finite 3-D integral, independent formulation and 2-D FEM (P118, P119, P121)	Historical 16.029 m/s is challenged; 12.448 m/s is an ideal screen, not a rated pass
Can a source supply the assumed shot?	Conditional voltage/current and bank cases (P120, P125)	No selected source or inverter; one commercial single-string ESR case **fails**
Does installed mass meet the predeclared 3U criterion?	Component rollup, 12-customer division and 15% parity band (P117, P123)	10.55 kg/3U exceeds the approximately 6 kg comparator and 6.9 kg allowed band; **fails**
Does the release yield mission value?	Historical-input orbit check and matched host/payload comparison (P115)	Immediate semimajor-axis result is conditional; no twelve-shot matched campaign closes

4.2 Reproducibility and uncertainty rule

Each quantitative result must retain its input configuration, model method, software/source file, output file, comparison band and stated limits. A second implementation under the same geometry is a useful **numerical cross-check**, but shared material laws and boundary assumptions remain common-mode uncertainty. Mesh or quadrature convergence establishes discretization behavior within a model; it cannot establish physical accuracy. Supplier ratings for the evaluated Gen5 launch-power chain, fabrication tolerance distributions, payload limits and host ICD limits are absent. Catalog feeder options used in FM08-R3 remain conditional prototype references. Deterministic gap, current and bank sweeps therefore show sensitivity, not measured statistical confidence intervals. The open issues in Appendix A are held as exceptions rather than treated as zeros.

4.3 Numerical checks and their limits

The magnetic field was compared across an analytical representation, magpylib, and 2-D/3-D finite-element calculations. A 3-D midgap FEM solve supports the field model at

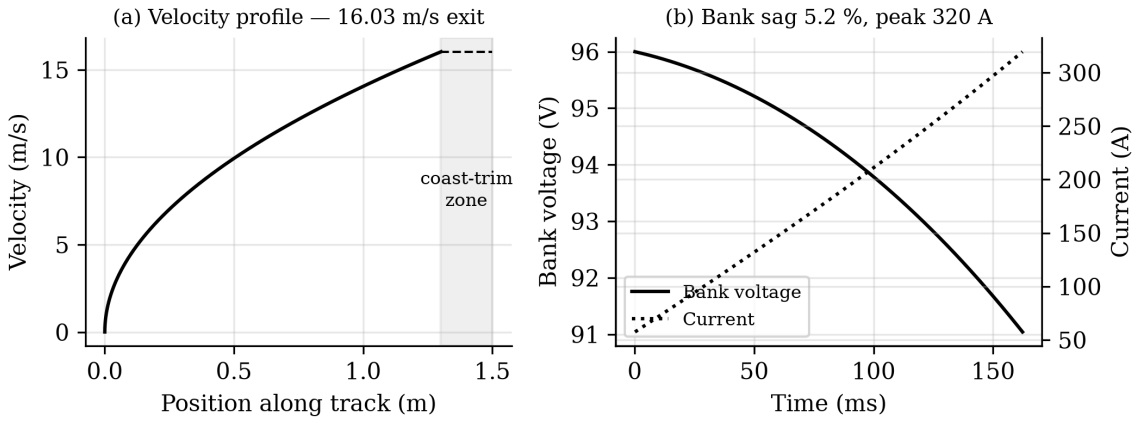


Figure 2: Historical modeled shot: velocity, bank voltage and current.

its tested location; P121 separately checks the ideal depth-integrated thrust by numerical surface-charge integration. Neither is a motor measurement. Shot and circuit predictions were compared with ngspice at stated assumptions. Structural calculations use CalculiX; airflow calculations use OpenFOAM. Orbital calculations include Cowell and GMAT cases. The independent orbit work rejected an earlier assertion that lifetime benefit was invariant across solar activity. Each check applies to its declared case and boundary conditions; none is a hardware test.

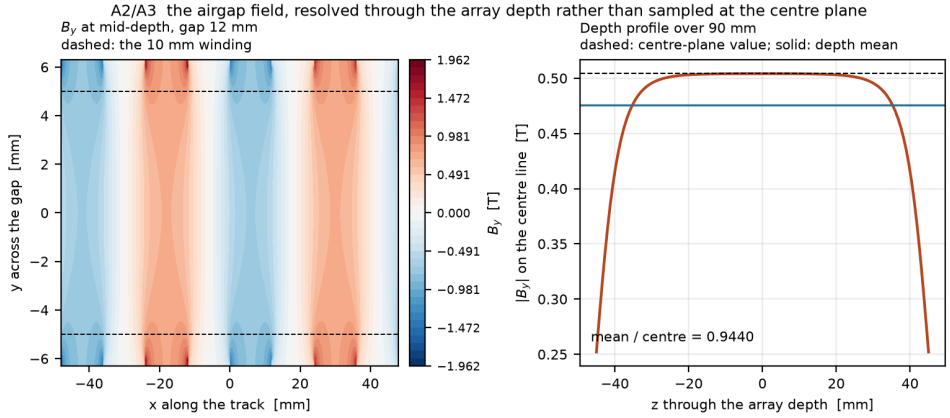


Figure 3: Representative magnetic-field model output. Source: thesis figure A02.

4.4 Governing relationships and boundary conditions

The historical shot model applies translational dynamics to the payload and reusable carrier. With moving mass m_{mov}, longitudinal electromagnetic force F_x, and represented resistance F_r,

$$m_{\text{mov}}\,\dot{v} = F_x(x, I, \phi) - F_r, \qquad \dot{x} = v.$$

The finite-force screen evaluates a winding-volume Lorentz integral and ideal longitudinal work,

$$F_x = \int_{V_{\text{Cu}}} (\mathbf{J} \times \mathbf{B})_x \, dV, \qquad W = \int_0^s F_x(x)\, dx, \qquad v_{\text{ideal}} = \sqrt{2W/m_{\text{mov}}}.$$

Optimal phase and the absence of electrical, switching and contact losses make this an ideal screen, not a component rating. The periodic-wave approximation $B(y) = B_0 e^{-ky}$, with $k = 2\pi/\lambda$, is distinguished from the finite cuboid-field and surface-charge implementations. The stator and mover end geometry therefore enter the decisive comparison explicitly [7,12].

A lumped winding/source model relates capacitor state and current through

$$L\,\frac{dI}{dt} = V_C - R_{\text{tot}} I - K_e v, \qquad C\,\frac{dV_C}{dt} = -I_C, \qquad Q_{\text{Cu}} = \int I^2 R_{\text{Cu}}\, dt.$$

These equations state the model family; the captured implementations retain their actual three-phase definitions and current conventions. A physically selected inverter, bus, winding, insulation and source are absent. Adiabatic $\Delta T = Q/(mc_p)$ is a single-pulse energy estimate and excludes heat transfer, hotspots and duty-cycle accumulation.

For two-body energy at an imposed release state,

$$\epsilon = \tfrac{1}{2}\|\mathbf{v}\|^2 - \mu/r, \qquad a = -\frac{\mu}{2\epsilon}.$$

For an internal release with pre-release system mass M, payload mass m, and specified payload-to-parent relative velocity **u**, momentum conservation gives

$$\mathbf{v}_{\text{payload}} = \mathbf{v}_0 + (1 - m/M)\mathbf{u}, \qquad \mathbf{v}_{\text{parent}} = \mathbf{v}_0 - (m/M)\mathbf{u}.$$

The point-mass host-burn calculation uses

$$M_{\text{after}} = M_{\text{before}} \exp[-\|\Delta\mathbf{v}\|/(I_{sp} g_0)].$$

The instantaneous two-body energy identity is not an atmospheric-life prediction. J2, atmosphere and host-control cases use separate inputs and boundaries [9]. Matched comparisons count device proxies, fuel, recoil and restoration at one terminal epoch. Ideal impulses do not resolve finite-burn navigation.

CHAPTER V

Controlled Gen5 configuration

The controlled Gen5 reference is the evaluated twelve-3U configuration. Dimensions and analytical revisions are held together even where they fail the intended criterion. The later R1 and Gen6 geometries are separate studies and cannot replace this reference without a new evaluation.

Table 5: Controlled Gen5 configuration — Parameter.

Parameter	Evaluated basis	Qualification status
Payload manifest	Twelve nominal 4 kg 3U spacecraft	Envelope/mass surrogates; no approved customer spacecraft
Host used for the matched Gen5 comparison	300 kg base host; 10 kg initial fuel; 220 s specific impulse	Computational surrogate; no selected provider
Structure	1.8 m longerons; release station 1.5 m from breech	Native/STEP reference retains interferences
Powered travel	1.3 m	Finite stator overlap decreases over the stroke
Carrier input	9.445 kg from historical Gen3 solid-volume accounting	Not a supplier-complete installed Gen5 mass extraction
Rail/cassette widths	205 mm and two × 166 mm; 526 mm available clear width	537 mm required; 11 mm shortfall
Electrical source	Modeled pulse bank and winding/circuit branches	No selected source/inverter/winding rating

5.1 Carrier, feed and release arrangement

The analysed system uses a double-sided Halbach-array linear synchronous motor to propel a reusable permanent-magnet sled to a release station 1.5 m from the breech on 1.8 m structural longerons. Two transverse cassettes are intended to feed twelve 3U satellites, but the present side-fed placement fails the fit check. A capacitor-based bank supplies the modeled pulse, and a brake is intended to arrest the sled after payload release. The **9.45 kg** sled input comes from historical Gen3 CAD solid volumes; this value reduced the exit speed relative to an earlier optimistic parametric estimate. The CAD render conveys a concept, not manufactured or assembled hardware.

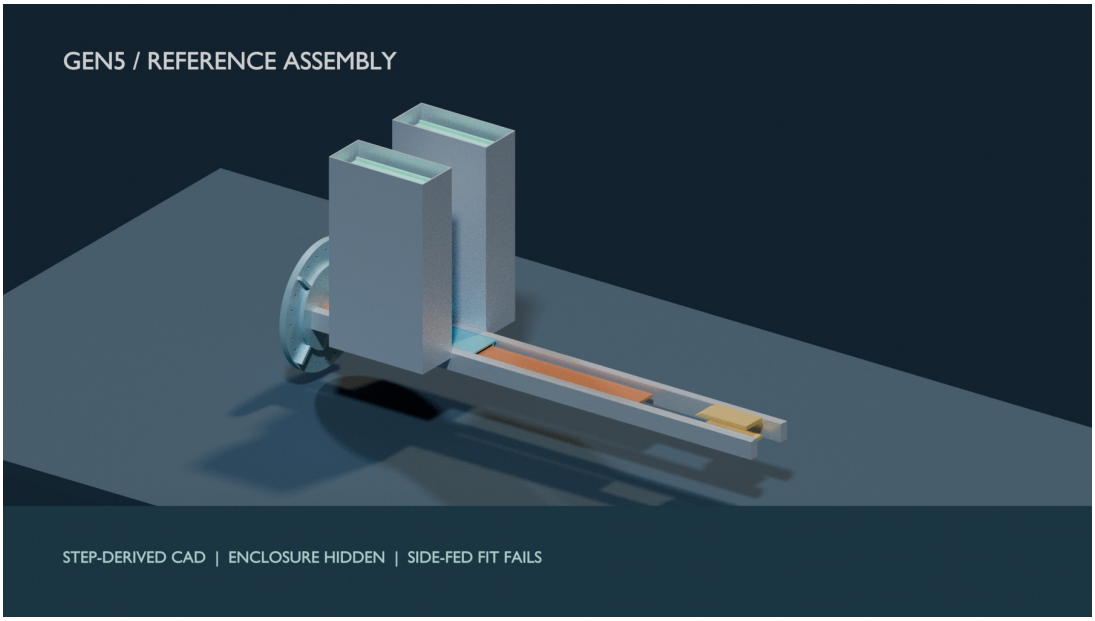


Figure 4: Gen5 reference assembly from STEP/FreeCAD rendered in Blender, with enclosure hidden. Recorded clashes remain; no physical article is shown.

CHAPTER VI

Electromagnetic drive, power and control analysis

6.1 Historical periodic model

The table below preserves the older model run for traceability. Its constant periodic-force input is challenged by the finite-geometry screen above. These numbers cannot be promoted together as the current machine's demonstrated or independently verified operating point.

Table 6: Historical periodic model — Model output.

Model output	Historical result	Interpretation
Exit velocity	16.029 m/s	Historical periodic-force input; finite geometry challenges it
Payload acceleration	10.07 g	Not a payload qualification claim
Pulse duration	162.3 ms	Modelled motor actuation
Peak current	320 A	Component implementation unverified
Electrical energy drawn, gross	2,782 J	Bank/circuit model
Recovered after release	47 J	39 mm available recovery zone
Sled energy to brake	1,162 J	Major loss and arrest burden
Electrical-to-payload efficiency, net	18.8%	514 J payload kinetic energy / net draw

The energy flow matters more than the velocity headline. The sled and its arrest hardware are part of the modeled price of avoiding a payload-side drive armature. Regeneration returns only a small fraction of sled energy within the available length. P117 separately checks the reported mass and energy identities and reconciles the historical **124.488 J (4.47%) compact-output remainder** to 90.964 J assumed converter loss, 32.460 J auxiliary draw

and 1.064 J of forward-Euler step excess, with a rounded residual below 0.001 J. This closes historical model arithmetic only. Selected hardware, switching and installed power verification remain open.

6.2 Finite geometry and conditional source cases

The 16.029 m/s shot result assumes periodic thrust throughout a 1.3 m powered stroke. A new 3-D analytic integration of the exact cuboid field over the CAD's finite 162-belt stator finds **1.042 kJ ideal work**, corresponding to **12.448 m/s only under optimal phase with switching, voltage and friction losses omitted**. The modeled mover array and stator stop directly overlapping after 1.066 m travel. The 336 mm magnetic model and 340 mm CAD envelope also disagree. Cross-section quadrature refinement changes the three checked station forces by at most 0.369%; halving station spacing changes work by 0.096%. The magnetic field law is shared with the earlier model, so this is an adverse analytic screen, **not an independent 3-D FEM or a hardware test**. The historical 16.029 m/s remains in older tables for traceability and must not be called demonstrated performance. See P118.

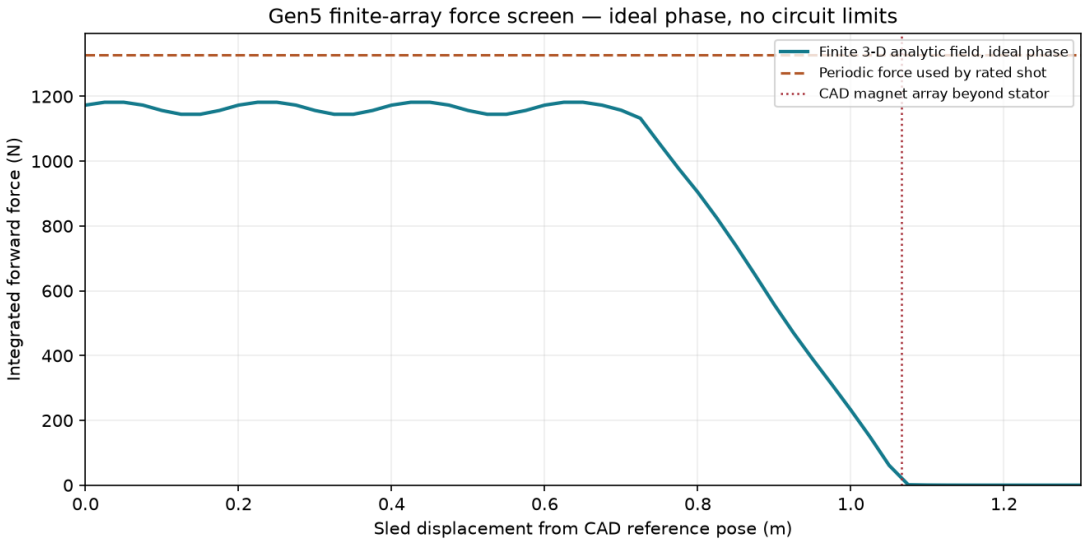


Figure 5: Finite-array/stator model force versus position; no physical measurement.

A separate **2-D finite-element magnetostatic solve** of the finite magnet array gives **1.082 kJ ideal work** on its finest 1 mm mesh, compared with the 3-D analytic screen's 1.042 kJ. The 2-to-1 mm mesh change is 0.59%; a larger air boundary changes the 2 mm work by 0.0004%. The solver uses a different field formulation but omits magnet-depth end effects, so the two absolute results must not be combined into an achieved speed. P119 records meshes, convergence and assumptions.

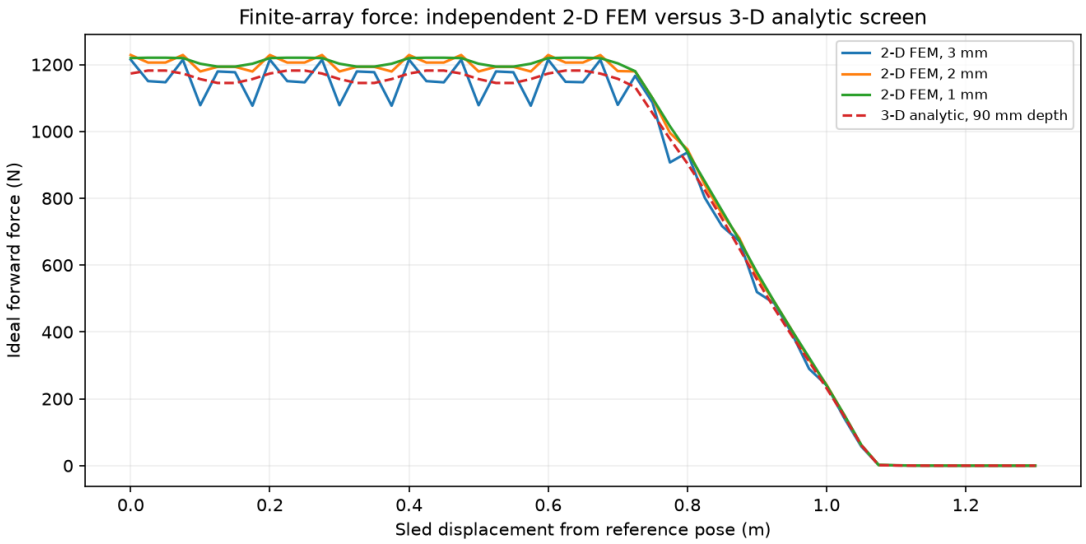


Figure 6: Independent 2-D finite-element force screen, not a measurement.

An independent **depth-resolved 3-D numerical surface-charge implementation** integrates all finite magnet faces and 162 belts. It returns **1.042 kJ ideal work**, agreeing with the analytic cuboid-field implementation under shared geometry, remanence and ideal-phase assumptions. Face quadrature refinement changes work by 0.00002%; halving station spacing changes it by 0.096%. This is a formulation cross-check, not 3-D FEM or hardware validation. An illustrative 14 mm face gap gives **0.907 kJ**, 13.0% below the 12 mm case. These deterministic cases are not fabrication tolerances. P121 and P122 include source, results and convergence.

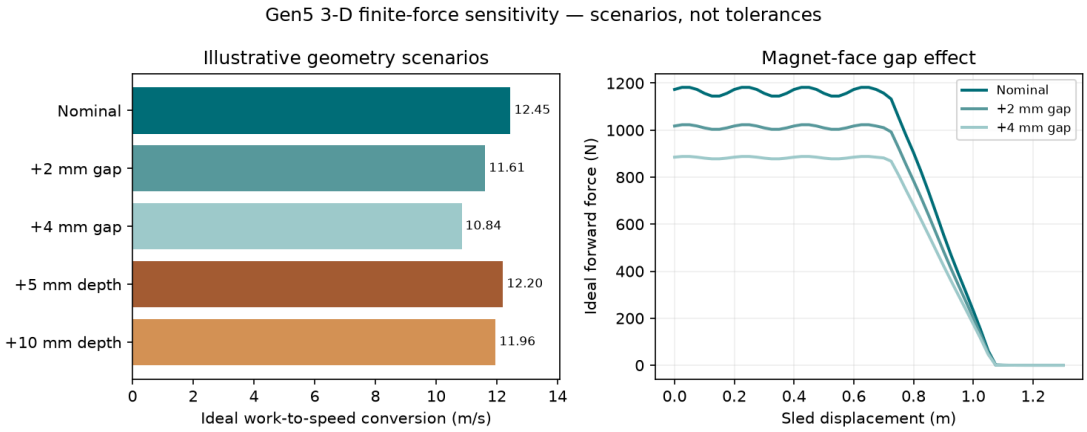


Figure 7: Ideal finite-force gap and depth scenarios; not measured tolerances.

A **conditional finite-force bank/trajectory rerun** retains the historical 96 V, 6 F, 12 mΩ source and assumed converter. With the full 1.3 m winding energized it computes **2.099 kJ gross capacitor draw** and 927 J copper heat; an illustrative 0.34 m active-copper branch gives 1.394 kJ gross and 243 J copper heat. Both report 12.448 m/s because the force curve

is imposed at ideal phase; no voltage-limited switching law or selected winding/inverter exists. The old 2.782 kJ, brake and control outputs are not revised ratings. P120 gives the time history and energy ledger.

The P125 conditional voltage/current screen adds a source-power check to the finite force map. Under the old, **unselected** 12 mΩ bank and periodic winding values, the quasi-steady reference still gives 12.448 m/s, while halving the phase-current limit gives 8.802 m/s. Under the older distributor-data 116–185 mΩ **single-string** ESR bound, the source-power quadratic fails at 0.612 m and 0.160 m respectively, before release. Ideal two- or three-parallel-string branches complete numerically only by multiplying capacitance, cell count and source mass. None is a manufacturer-rated, thermally checked switching design; the model assumes instantaneous q-axis current. The 12.448 m/s result must therefore not be presented as a sourceable release rating.

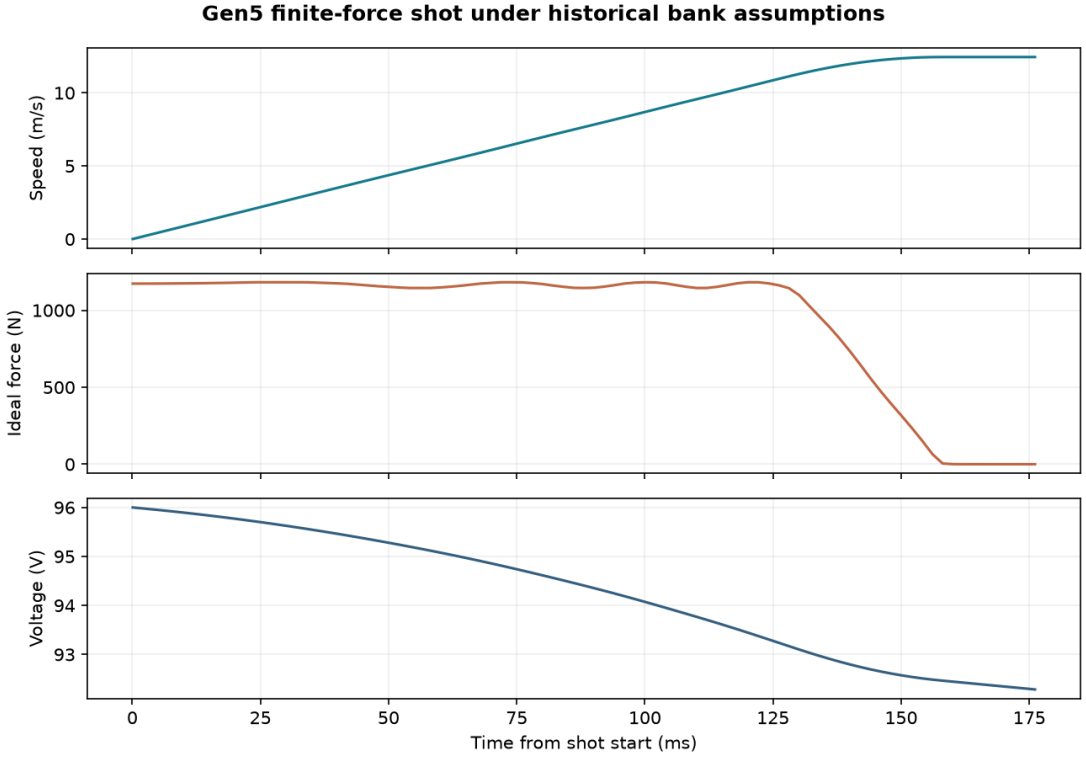


Figure 8: Finite-force speed and assumed bank energy histories.

6.3 Historical control simulation

The closed-loop Monte Carlo model predicts **0.0274 m/s (3σ)** exit-velocity dispersion around a **15.8 m/s** fleet setpoint. It uses assumed sensing and plant tolerances, with

headroom below the open-loop ceiling. This is a simulated distribution, not a measured release repeatability.

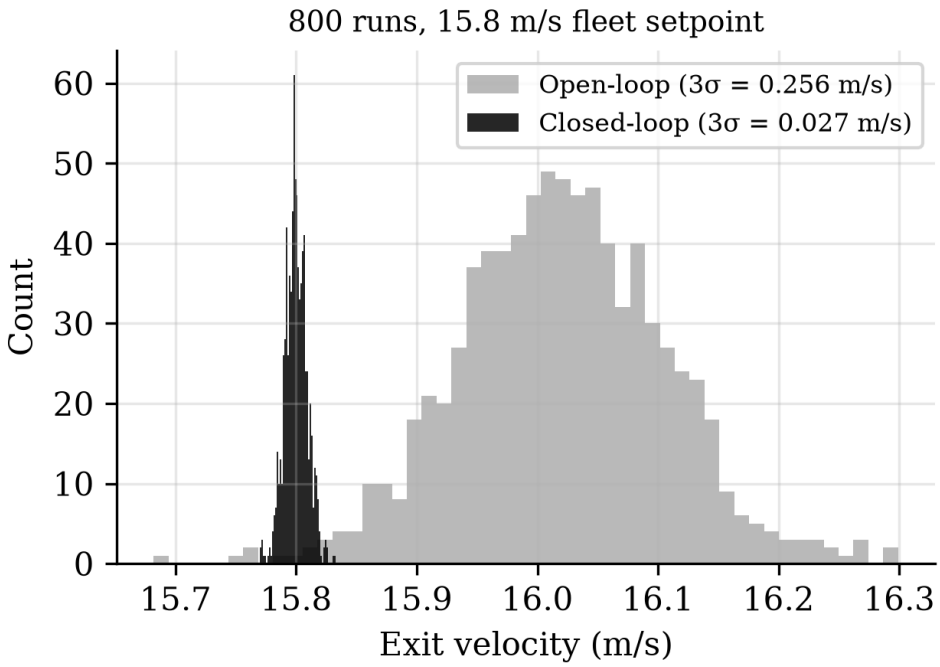


Figure 9: Closed-loop simulation distribution at the fleet setpoint.

CHAPTER VII

CAD integration, packaging and installed mass

7.1 Installed mass and comparator

The modeled Gen5 dry rollup is **126.6 kg** and loaded mass **174.6 kg**; these include assumed hardware and historical Gen3 sled volume, not a mass extraction from a complete installed Gen5 assembly. At twelve 3U customers, the deployer weighs **10.55 kg per customer**, compared with approximately **6 kg** for the spring canister comparator used in the analysis. The ratio is **1.758**; the preset acceptance band required parity within 15%. That band fails by a large margin. Replacing an **8 kg enclosure placeholder** with a **50.04 kg** geometry-derived buildup exposed much of the penalty. Even changing structure alone does not recover parity in the studied options. The result is a reason to reopen the architecture, not to conceal the comparison.

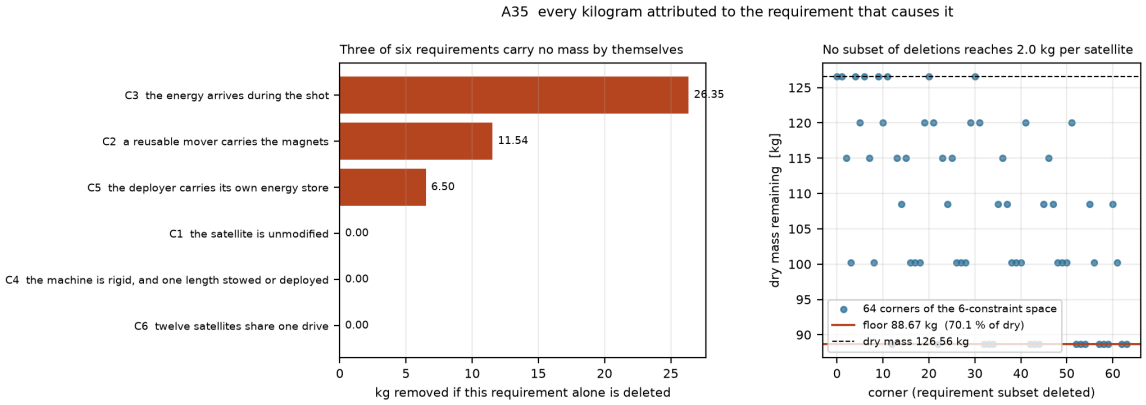


Figure 10: Constraint and mass-attribution analysis; mass is a model rollup.

7.2 Native solid and assembly checks

The geometry package now includes a FreeCAD 1.0 native document, eight re-exported STEP parts, and a 20-instance STEP assembly. Each imported solid passed FreeCAD validity and nonzero-volume checks. The source solids were produced in CadQuery; importing them into FreeCAD does not create parametric feature history or prove that the

assembly can be manufactured. The CAD review report identifies source revisions and checks, while P116 records the numerical intersection result.

The side-fed reference assembly has **526 mm** clear enclosure width. Its **205 mm** track plus two **166 mm** cassettes require **537 mm**, an **11 mm** shortfall even before running clearance. The exact solids overlap by **32,915 mm³ per side** in this placement. A full census of the 20 leaf solids finds **17 intersecting pairs**. In the reference placement, the sled intersects the track by **82,983 mm³** and stator by **6,300 mm³**. The interface-ring proxy also intersects track, stator and enclosure proxies; stored-payload boxes intersect cassette bodies. Proxy contact and physical collision require separate engineering disposition, but the sled/track and sled/stator conflicts prevent acceptance of the assembly as drawn. The scripted feeder-path check did not evaluate these pairs. A different feed architecture, enclosure or dimension set requires a new configuration and reassessment; the current render must not be presented as an assembled flight article. The all-leaf audit supplies the pair inventory and source script.

7.3 Separately evaluated widened geometry and arrest bounds

An **unselected R1 geometry candidate** uses a 570 mm wide enclosure. Its native FreeCAD document and STEP parts clear the original cassette–track overlap and twelve scripted 3U envelope transfer paths against selected fixed parts. A subsequent all-leaf exact-solid audit finds five other intersecting pairs in this widened assembly, including sled–track and sled–stator conflicts. The reference track beam occupies the roller corridor because its solid model has no channel cut; the sled backstop occupies the stator proxy. R1 therefore does not pass complete assembly integration. It also lacks the actual lift/carriage, launch restraint, actuation, tolerances, fault recovery and revised installed mass. Its increased envelope has no approved host. This candidate does not alter the evaluated Gen5 configuration or repair the performance discrepancy. See R1 CAD record and the all-leaf audit.

The separate P126 lateral tolerance screen uses R1's CAD-derived 5 mm nominal cassette-to-track gap. With declared 1 mm track placement, 1 mm cassette placement, 0.5 mm combined form and 0.2° angular error across a 340.5 mm payload proxy, worst-case residual is 1.31 mm; a 0.5° band overlaps by 0.47 mm. These are illustrative assumptions, not measured process capability or released GD&T.

P127 conservation bounds put 731.8 J of kinetic energy into the modeled 9.445 kg sled at the **ideal** finite-force speed. Across the provisional CAD brake's 210 mm length, stopping

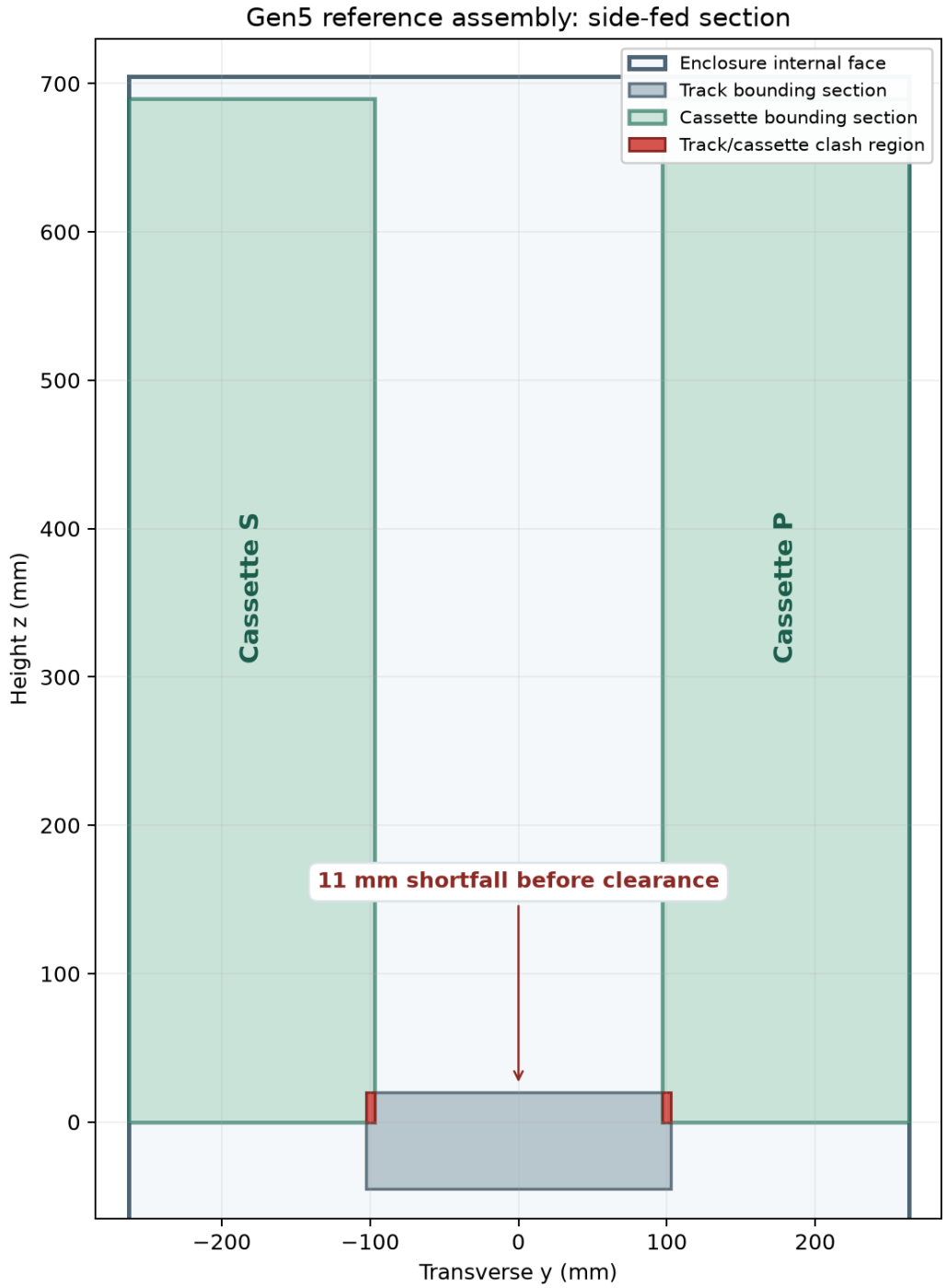


Figure 11: Dimensioned side-fed reference cross-section; this configuration fails the width check.

requires at least 37.6 g mean sled deceleration. Eddy-current peak force, repeated-shot heat, payload contact and host attitude remain unsolved; the mean bound is not a brake acceptance test.

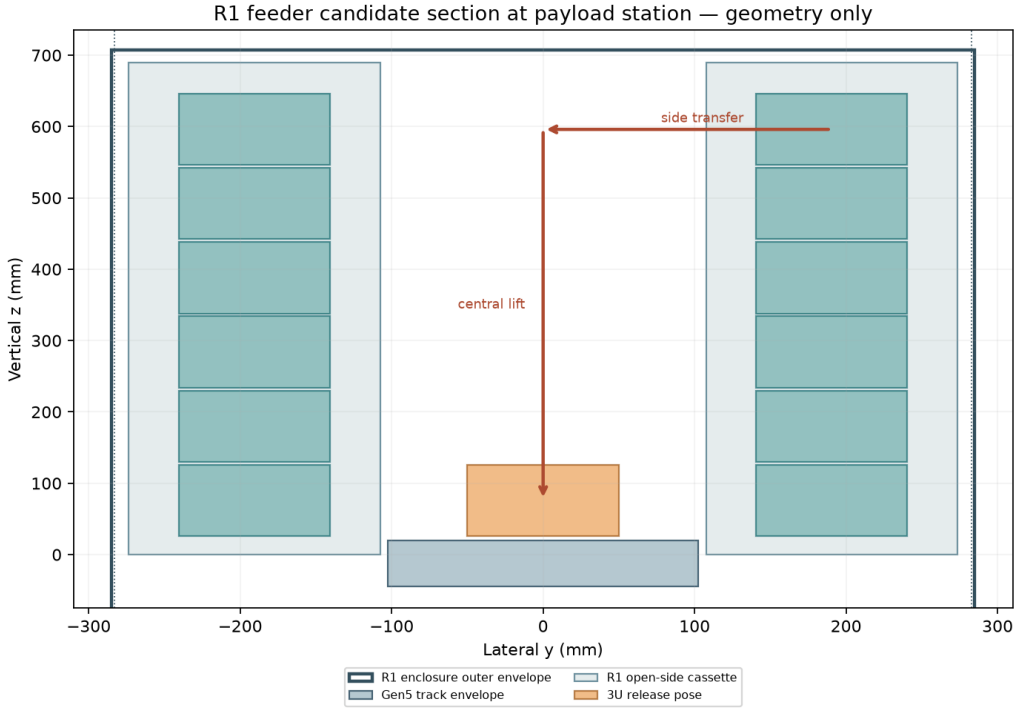


Figure 12: R1 candidate section; geometry drawing, not a qualified mechanism.

7.4 Mass-property scope

The 23-line historical mass model sums to 126.561588 kg dry, with an assumed axial CG of 0.457690 m. This is not a complete Gen5 material extraction or full inertia tensor. Potential enclosure/thermal/bracket allocation overlap remains unresolved; no unverified mass credit is taken. The original mass failure is retained.

All 17 evaluated Gen5 and five widened-R1 intersections are classified as confirmed conflicts or unresolved adapter/payload proxy intersections. None is accepted as intentional contact without an interface definition; no geometry repair is claimed. Historical isolated FE results do not qualify the failed assembly.

7.5 Selected Gen6 feeder architecture

The separately controlled **FM08-R3 feeder** uses fixed dedicated cells with guided lead-screw pushers and one indexed receiver. A fixed spacer accommodates each 1U payload

without modifying the satellite or launching the spacer. A continuous capped channel contains the payload during transfer. The receiver far stop is proved closed before the cassette gate opens; seated-presence and settling proof precede near-side receiver capture and pusher return. Independent cell actuation supports isolation of a faulty cell, subject to a verified clear common path.

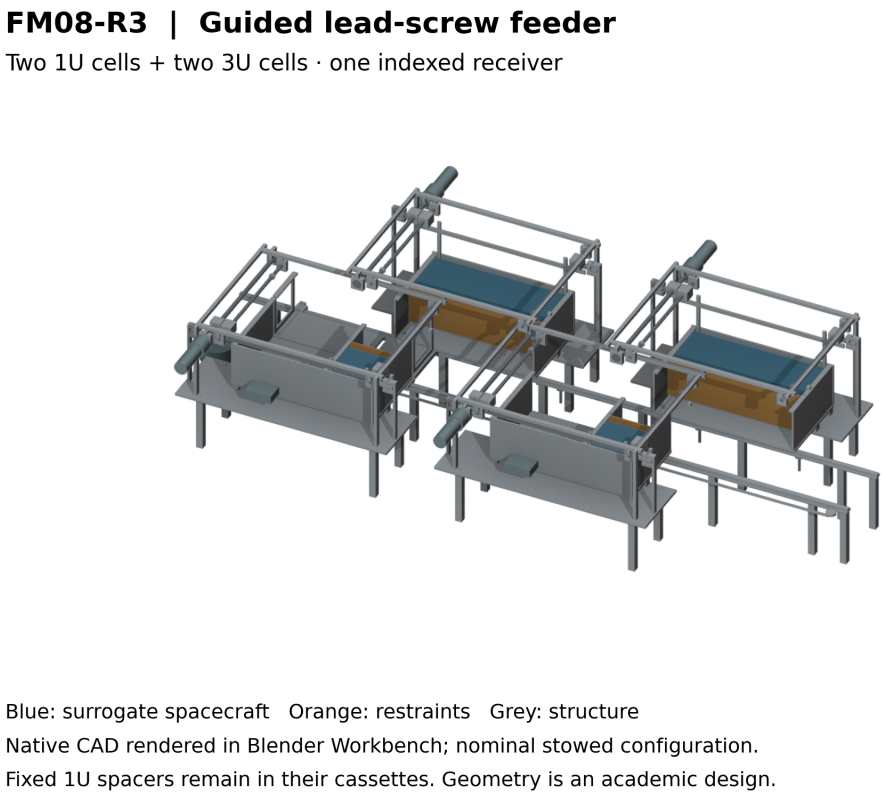


Figure 13: FM08-R3 selected four-cell assembly. Visualization derived from native CAD; shown geometry and catalog envelopes describe the computational design, rather than manufactured hardware.

This selection uses an explicitly revised academic design criterion: a justified architecture trade, editable CAD and STEP, nominal operating-envelope checks, declared tolerance and analytical load screens, catalog-based actuator sizing, and executable sequence/fault verification. Architecture selection is supported by verification of the selected configuration; the alternative concepts retain their dated comparative evidence and research dispositions. It does not convert CAD sampling into manufacturing tolerance verification or a software interlock model into demonstrated hardware reliability.

The selected feeder package contains the design record and the reproducible acceptance evidence. Appendix C reports the configuration, calculated results and remaining physical validation boundary. The frozen Gen5 reference and its evaluated mass, fit and motor results remain unchanged.

CHAPTER VIII

Orbital mechanics and matched mission comparison

8.1 Immediate orbit and atmosphere-model lifetime

For the stated orbit and mean-solar-activity case, an **assumed historical 16.029 m/s release** predicts a **28.8 km** semi-major-axis increase and a **1.60×** lifetime multiplier for a propulsionless satellite. A separate Cartesian two-body integration of that assumed historical release state returns **28.800775 km**, agreeing within the declared numerical band. This checks only the immediate orbital-energy result. P134 independently integrates the shared static-atmosphere model and gives a 1.602165 ratio; that numerical agreement does not validate real lifetime or solar invariance. Timing alone, with zero relative impulse and no host or drag difference, produces neither persistent phasing nor this semi-major-axis change.

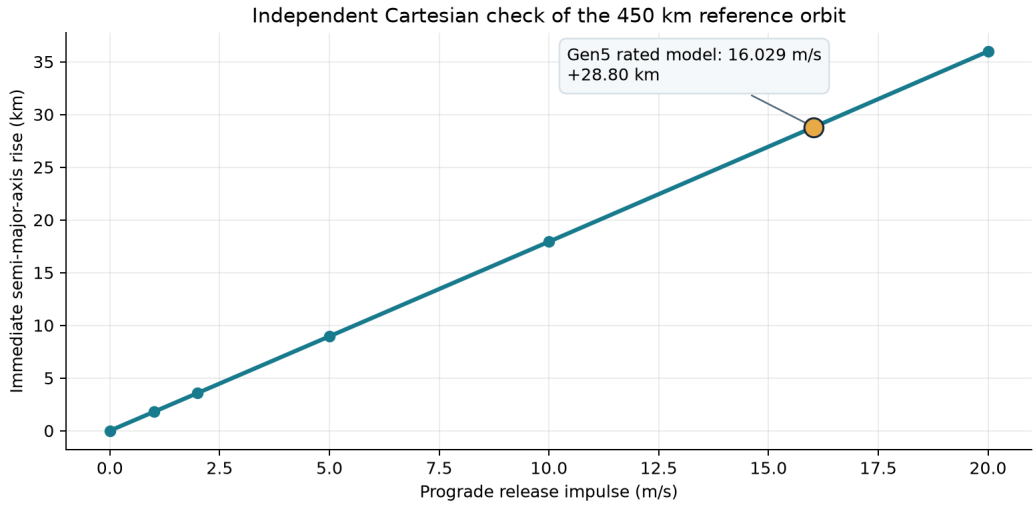


Figure 14: Independent rated-state two-body orbit check. The lifetime claim is outside its scope.

P134 uses a separate orbit-averaged DOP853 integration of the shared static atmosphere. At imposed inertial increments of 2.5, 12.448437 and 16.029 m/s it gives lifetime ratios of 1.082110, 1.451222 and 1.602165. The maximum refinement change is 0.000145%, below its 0.5% numerical band. Space weather, ballistic coefficient history and actual achieved release remain outside that evidence.

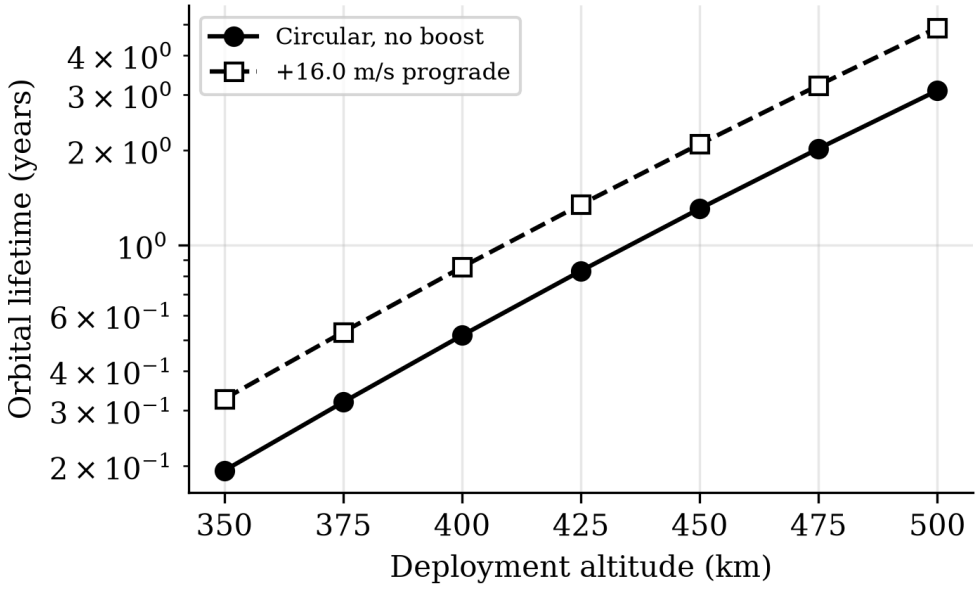


Figure 15: Modelled orbital-lifetime response. Absolute values depend on atmosphere.

8.2 Matched Gen5 host and payload comparison

A **matched reference comparison**uses twelve identical 4 kg payloads at 450 km, the same 300 kg base host with 10 kg assumed fuel and 220 s specific impulse, and installed device masses of 72 kg for a spring class versus 126.562 kg modeled Gen5. To meet one common +16.029 m/s inertial payload target at the same epoch, ideal host preburn takes 2.693 kg of propellant with 2.5 m/s spring release and 0.827 kg with the 12.448 m/s finite Gen5 estimate. The subsequent common-input, assumed-100 N twelve-shot optimizer accepts prefixes of **4/12 spring, 1/12 finite Gen5, 1/12 historical Gen5**. None closes. These are bounded numerical cases, not a supplier quote, proof of infeasibility or product advantage. Onboard propulsion, transport services and long-horizon drag remain unmodeled. See matched mission inputs.

8.3 Service precision and scaling bounds

The economic case is a **delivered orbital state at a specified epoch and tolerance**, not a high release speed by itself. P123's reproducible one-event mass screen finds that, with the same assumed host and 10 kg initial fuel, an ideal finite-force Gen5 release saves 1.866 kg of host propellant relative to a 2.5 m/s spring release. Its modeled hardware is 54.562 kg heavier. If fuel is resized to the one event plus the same 2 kg reserve, Gen5 still carries 52.717 kg more device plus fuel. Even a hypothetical zero-burn release needs a device

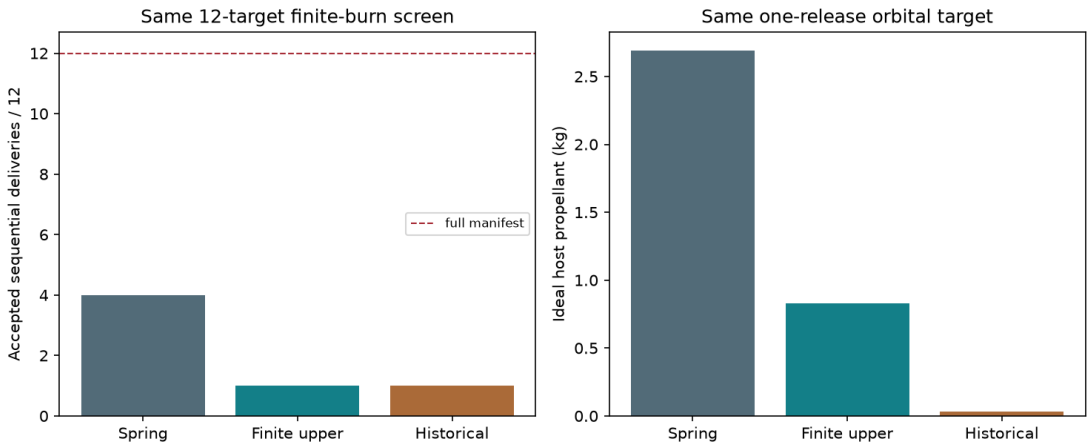


Figure 16: Matched reference results; assumed host and device classes.

at or below 74.659 kg to tie this specific spring reference. This is a serious architecture requirement, although it is not a full twelve-shot economic result or a price for a tailored orbital state.

P124's independent orbit and stroke screen works backward from ideal orbit targets. At 450 km circular altitude, a 20 km immediate semi-major-axis rise requires about 11.149 m/s of *inertial tangential* increment. A ±0.1 km axis band leaves roughly 0.0555 m/s on its smaller side **if every other error is zero**. A flight requirement must divide the error among host state, release direction and speed, timing, orbit propagation and tracking, and must include along-track, plane and disposal constraints. The old 0.0274 m/s modeled release dispersion is unmeasured and cannot be applied to P118's challenged finite-stator case.

Over the evaluated 1.3 m powered stroke, an ideal 10 g constant-acceleration limit allows 15.97 m/s. At 1 km/s, the same stroke implies roughly 39,220 g and 2 MJ of payload kinetic energy for a 4 kg 3U; a 10 g stroke would be about 5.10 km. This makes the 1 km/s objective unsuitable as the specification for a first ordinary-3U Gen6 prototype. The first test article should instead inherit a **named mission, payload load limit and host interface**, and select a lower-speed command range only after a matched service comparison. This preserves the longer-range research goal while making prototype tests technically meaningful.

8.4 Native propagation harmonization

The local harmonization report isolates the earlier gravity-axis/configuration mismatch. GMAT R2026a and Orekit 13.1.6 agree to approximately **1.285 mm maximum** over one day with shared prescribed pole and JGM2 constants, using one continuous GMAT arc. Independent Cartesian and time-step comparisons are retained. This is numerical

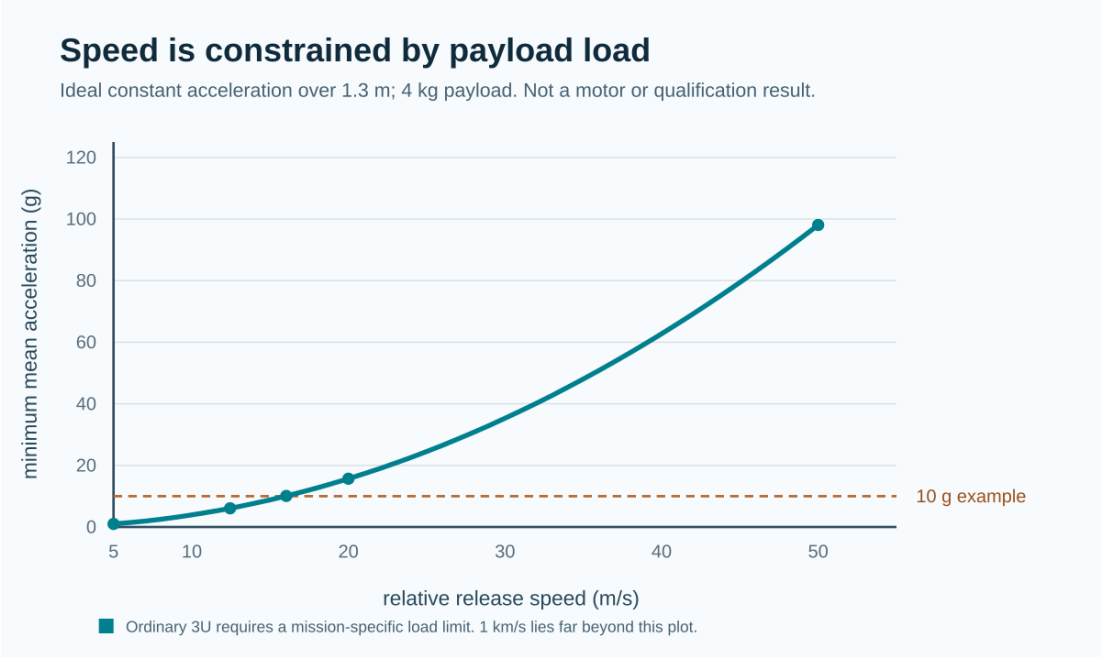
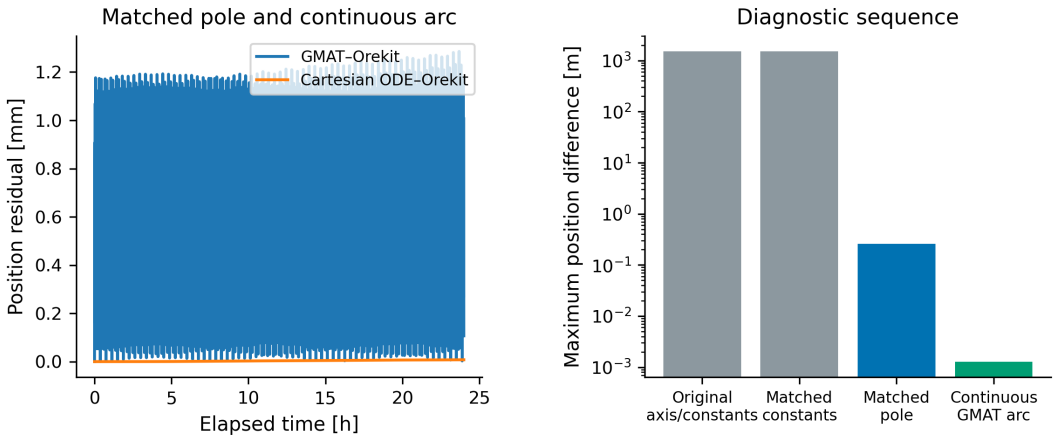


Figure 17: Ideal acceleration required by speed on a 1.3 m stroke; not a VOLLEY motor rating.

consistency, not physical orbit accuracy, independent Earth-orientation validation or demonstrated release speed. The frozen Gen5 record is unchanged.



450 km, 51.6°; one day, J2 only. Shared prescribed GMAT EarthFixed pole. First three bars: SciPy versus repeated-stop GMAT; last: Orekit versus continuous GMAT. Different numerical histories are declared; no flight-accuracy or motor validation.

Figure 18: Matched native solver result and diagnostic sequence; numerical model agreement only.

8.5 Historical multi-orbit archive

The corrected A15 R1 archive, executed in GMAT R2026a on 30 September and reparsed on 16 October, returns a 14.558545° circular RAAN span and a 190.149944 m focused-

window minimum sampled separation. These preloaded orbits do not establish continuous clearance or a coupled sequential campaign. The historical 367° span is withdrawn.

CHAPTER IX

Results and engineering discussion

The evaluated configuration has three decisive outcomes: its mass criterion fails, its side-fed packaging criterion fails, and its historical speed claim is challenged by finite geometry. The result is a negative design disposition for this particular twelve-3U Gen5 reference. It does not rule out all commanded-deployment concepts.

Table 7: Results and engineering discussion — Question.

Question	Result	Decision
Does Gen5 meet the stated 3U mass-parity band?	10.547 kg per customer versus a 6.9 kg allowed comparison band	Fails the model criterion
Do the reference parts coexist with positive clearance?	11 mm width shortfall and 17 exact-solid intersecting pairs	Fails the represented geometry
Is 16.029 m/s supported as an installed rating?	Finite geometry gives 1.042 kJ ideal work and a 12.448 m/s ideal upper screen	Historical prediction challenged; no rating selected
Does the tested twelve-release Gen5 campaign close?	All sampled optimizer cases reject complete delivery	No closure demonstrated; not a universal impossibility proof
Is physical release quality established?	No VOLLEY prototype or measured releases	Unverified
Does later native propagation agreement prove precision?	Same-model numerical consistency under prescribed controls	Software verification only

The analytical result has four parts. The historical model represents a proposed commandable release and a conditional orbital-energy benefit, but its speed is challenged by finite geometry; the full-system 3U mass criterion fails; the stated side-fed layout fails its CAD width check; and physical behavior remains unverified. The latter includes

release-cycle reliability, shock and arrest loads transmitted to stowed satellites, field exposure near a customer payload, host attitude/control authority and provider approval. The known-problems register ranks several as potentially design-fatal. These findings limit product claims and define the research conclusion.

The field, circuit, CAD, mass and mission studies are not interchangeable demonstrations. A converged numerical answer under a shared assumption can coexist with an incorrect physical assumption. Similarly, nominal nonpenetration of a simplified geometry can coexist with an unretained payload after power loss. The study therefore retains failed cases and open mechanisms rather than interpreting coverage as acceptance.

CHAPTER X

Computational resources and limitations

10.1 Computational resources

The work uses inspectable source models and captured outputs. Python provides analytical dynamics, quadrature, interval and sensitivity calculations; FreeCAD/OpenCascade provide native solid and STEP inspection; Blender supplies explicitly labelled renders. Native GMAT R2026a and Orekit 13.1.6 provide the stated orbital cases. CalculiX fixture results apply only to their declared support and loading. Historical OpenFOAM, ngspice and electromagnetic records have case-specific run sheets; they are not blanket verification of a current installed assembly.

No physical mechanism was fabricated, weighed, released or environmentally qualified. The report contains no measured VOLLEY reliability, magnetic cleanliness, shock or deployment-repeatability result. Public hardware and published tests support methods and comparisons only.

10.2 Technical limitations and retained exceptions

Gen5 is a computational study with recorded adverse findings. Its evaluated configuration is not a qualified deployer or released product. The decisive electromagnetic, feed/release, brake, host, structural and installed-system questions remain open or bounded as recorded in the validation files. Physical bench and qualification tests are future work and remain marked unrun.

Table 8: Technical limitations and retained exceptions — System question.

System question	What this study actually establishes	Final Gen5 decision boundary
Motor and electrical source	Historical 16.029 m/s periodic-model shot; finite-geometry ideal-phase screen 12.448 m/s; historical gross/net shot energy reconciled algebraically in P117	**No component-rated release speed.** Winding, switching, source impedance, derating and coupled repeated-shot performance are unselected. The historical value is a model output, not a flight command.
Shot and brake heat	P117 identifies modeled copper, bank, converter and auxiliary terms; P127 puts 731.8 J into one ideal sled arrest and 8.782 kJ across twelve identical events before recovery/cooling	Heat deposition, peak temperature, duty interval, material limits, source warming and brake cooling are **unverified**. The twelve-shot sum is an energy bound, not a thermal qualification.
Magazine feed and release contact	The evaluated cassette placement fails by 11 mm; R1 gives a separate scripted path; tip-off calculations are model studies	No selected Gen5 feed actuator, launch restraint, contact law, tip-off acceptance or recontact test exists. No verified release-quality or P-POD-class reliability claim is made.
Carrier arrest and return	P127 gives a 3.485 kN *mean* braking-force floor over the provisional 210 mm corridor at the ideal 12.448 m/s screen	Peak force, brake heating, carrier containment, safe stop, reset, jam and abort behavior are **unverified**. Mean deceleration does not prove peak deceleration.

System question	What this study actually establishes	Final Gen5 decision boundary
Structure and environment	Preliminary structural and geometry studies identify loads and the side-fed clash	No same-revision launch-load, vibration, shock, thermal-distortion or converged structural acceptance is claimed.
Magnetic/payload compatibility	Field studies identify exposure concerns, including possible magnetometer saturation	A passive spacecraft is only a surrogate envelope. Payload magnetic limits, shielding, safe standoff and provider approval are **unknown**.
Orbit and campaign	Independent Cartesian two-body calculation checks only the immediate 28.800775 km axis change *if* the historical 16.029 m/s departure state is imposed; sampled twelve-shot cases do not close	The 1.60× lifetime is a conditional atmosphere-model output; P134 checks its numerical implementation under shared assumptions, not the real atmosphere. Delivered-state covariance, conjunction clearance, host abort/recovery and disposal are outside the demonstrated mission result.
Installed system and interface	Modeled 10.547 kg per carried 3U fails the study's mass criterion; side-fed reference fails fit	No supplier-complete installed bill of materials, named accepted spacecraft/host/provider interface or packaging approval exists.

In this report, **failed** denotes an evaluated configuration that misses a declared criterion; **unverified** denotes an effect not resolved by the reported method. Neither status changes

because a separate Gen6 geometry has been explored. The one-cell feeder study retains an unresolved power-loss restraint condition and does not establish a working Gen5 magazine.

10.3 Required physical correlation

Table 9: Required physical correlation — Future validation.

Future validation	Purpose	Current status
Instrumented force/current/position tests	Correlate force map, commutation and winding/source model	Unrun
Repeated restrained transfer and fault injection	Establish actual custody, gate holding, jam/abort/reset behavior	Unrun
Dummy payload release/contact tests	Measure velocity, tip-off, recontact and payload loads	Unrun
Instrumented carrier arrest and thermal cycling	Establish peak load, containment, heat and repeated-shot duty	Unrun
Vibration, shock and thermal-vacuum tests	Verify an actual manufactured configuration against its provider limits	Unrun; provider limits unselected
EMC and magnetic compatibility	Establish payload-specific electrical/magnetic accommodation	Unrun; payload/interface limits unselected

Simulation cannot supply the acceptance of a launch provider, a calibrated hardware result or a reliability demonstration. Those requirements remain conditions for future hardware development. No safety or qualification pass is claimed from the analytical fixtures.

CHAPTER XI

Conclusion and future development

The Gen5 computational thesis reaches a research conclusion within its stated scope. Its historical model predicts a moderate-acceleration release, but finite geometry challenges the speed; an independent two-body calculation checks only the immediate orbital change under that assumed release state. Installed-mass accounting misses its preset 3U benchmark, and the side-fed reference CAD assembly fails its width check. The current configuration is therefore not a selected flight baseline. Hardware behavior and interfaces remain outside the tested evidence.

The thesis's strongest engineering contribution is the transparent decision record: system budgets and idealized mechanisms were modelled, independent numerical checks corrected several claims, and an unfavourable mass result was retained. That closes the Gen5 study and provides a defensible input to the separately controlled Gen6 design study.

The separate FM08-R3 work adds a selected mechanical subsystem: guided pusher transfer from dedicated mixed-size cells into a reusable receiver, with retained spacers, capped-channel containment and explicit boundary handover, conditional actuator sizing and executable interlock tests. This provides a defined engineering basis for a future instrumented article. The selected subsystem does not establish the launch motor's attainable speed, a whole-system economic advantage or physical reliability.

11.1 Gen6 direction and research boundary

Gen6's controlling objective is a reusable launch path that sequentially accepts different supported CubeSat sizes and commands each departure condition. The approximately 1–2 m/s to 1 km/s span is an investigation envelope; the upper end may force acceleration, energy, travel, payload or provider constraints that reject candidates. Electromagnetic systems are preferred for comparison but are not preselected. Competent springs, electromechanical/stored-energy, gas/fluid, hybrid and conventional mission substitutes belong in the same comparison boundary.

The next milestone is an independently reviewable test-article configuration: controlled CAD and drawings; a sourceable bill of materials; host/payload interfaces; installed mass, energy and thermal accounting; assembly and inspection procedures; instrumented tests with predetermined pass/fail and stop criteria; uncertainty and retained failures; and an explicit build/test release decision. A rejected candidate remains a valid research result.

Appendix C details the passive 1U/3U service objective, manoeuvring-host surrogate and selected FM08-R3 feeder design. Its 3–6 m/s first-article objective and distant 1 km/s research objective are distinct. The feeder is selected at the computational academic-design level; physical performance, environmental qualification and a hardware build release are not asserted.

APPENDIX A

Numerical and CAD evidence trail

Table 10: Numerical and CAD evidence trail — Result.

Result	Reproducible record	What remains outside the check
Historical periodic shot and energy	`analysis/results/motor_results.json`, `analysis/motor_model.py`	Unmeasured force and actual circuit parasitics
Imposed-state immediate orbital change	`P115`, `rated_orbit_independent.py`	Atmospheric lifetime and controlled host mission
3U mass decision	`mass_properties.json`, `BASELINE`	Hardware weighing, complete installed host burden
Native geometry and collision	`Gen5_Review.FCStd`, `FREECAD_EXPORT.json`, `P116`	Tolerance, feed motion, payload accommodation and provider fit
Wider model and open gates	`computational review`, `CAD review`, `open problems`	Final verification closure and physical development

The run sheets report software, inputs, declared comparison bands and files. Figures are generated model or CAD visualizations. A screenshot of a solver interface would not by itself establish a correct result; the editable model, data and case boundary are the audit trail.

A.1 Current technical-evidence identity

The unchanged 68-file Gen5 technical baseline controls the evaluated physics/CAD record. The numerical checks and later Gen6 studies are identified separately. The submission identity file records the exact source, figure, PDF and DOCX hashes for this issue; this is a document freeze, not a hardware release.

Table 11: Current technical-evidence identity — Evidence package.

Evidence package	Local source and captured results	Claim boundary
Gen5 recorded failures	Engineering audit	Computational criteria and open exceptions
GMAT/Orekit harmonization	Native propagation	Shared prescribed pole and gravity; software consistency
Four/twelve campaign construction	Coupled delivery	Constructed targets, proxies and ideal impulses
Seven feeder concepts	FDR-08	Comparable bounded screens; no hardware downselect
Captive-transfer correction	CT-01	Prescribed primitives; actual custody/contact open
Replenishment concept	RPL-01	Negative baseline integration disposition
Full native event replay	CNR-01	Numerical cross-check and assumed error budgets

APPENDIX B

Development history

B.1 Documented evolution

Table 12: Documented evolution — Period.

Period	Milestone	Confidence in date
2023	Free-flyer idea reframed around a host platform	Documented design decision
Mid-2025	Coilgun concept shifted toward a linear synchronous motor	Approximate historical milestone
2025–July 2026	CAD generations and mass model developed	Some dates reconstructed/approximate
July–September 2026	Gen5 analysis, numerical checks, defects and thesis record	Dated repository/analysis records
October 2026	Gen5 findings and design decision assembled for academic review; final presentation scheduled for 31 October	Review package dated 10 October

The thesis repository's reconstructed history expressly marks several early milestone dates as approximate. The analytical baseline was revised as evidence improved: a CAD-derived sled mass displaced a lighter parametric estimate; a depth-resolved thrust calculation lowered the historical periodic-model speed; an enclosure buildup exposed the 3U mass penalty; and an independent orbit check rejected an overbroad solar-invariance claim. These revisions are part of the review record; the historical 16.029 m/s shot remains traceable, but no final speed rating is selected. Gen6 studies are separately identified in Appendix C.

APPENDIX C

Separate Gen6 research studies

C.1 Surrogate service and first-article progression

(P130–P132)

The follow-on mission is an explicitly **surrogate** four-release case: two passive 1U and two passive 3U spacecraft, a roughly 500 km sun-synchronous host, all releases within one orbit, and a distinct full position/velocity target for each satellite 24 hours after the last release. The matched J2 campaign lets both commanded release and spring plus host maneuvers meet the **same** four exact mathematical targets. With assumed installed masses of 30 kg for VOLLEY and 16 kg for springs, VOLLEY saves 3.2823 kg of host fuel but adds 14 kg of installed device, leaving a 10.7177 kg penalty on the narrow device-plus-consumed-fuel measure. A +0.1 m/s release-speed perturbation moves the one-day terminal position by roughly 25–27 km in either strategy. Exact numerical matching is not delivered-state accuracy; the study has no customer tolerance, navigation covariance or physical dispenser rating.

Two editable FreeCAD/STEP 3U first-article envelope candidates provide a separate geometry screen. A captured permanent-magnet carrier is the analysis lead; a captured conductive-plate carrier remains a challenger. Their **partial** frame, guide and carrier material proxies are 9.926 and 9.945 kg. Under the surrogate P130 mission, only about 9.35 kg remains for all unmodelled hardware before device-plus-consumed-fuel parity is lost. Five nominal positions have no solid intersection, but the assumed 0.7° yaw with a 1.5 mm linear stack clashes with the simple guide corridor. P131 gives an ideal 6 m/s moving-energy floor of about 82 J. At an assumed 10% wall-to-kinetic efficiency, P132 requires at least 818 J of input; it does not select a source, motor or brake. P130 used hypothetical release commands up to approximately 9 m/s, outside the proposed 3–6 m/s first-article band. The later coupled construction below limits the imposed release input to 3–6 m/s and supplies the remaining change through host assistance. Neither policy establishes actual motor performance.

A separate one-cell fixed-cassette feeder study begins the transfer design with one inert 3U envelope and a captive lead-screw pusher as an analysis reference. Four editable FreeCAD

states and STEP exports depict stowed, armed, midway and seated positions. Its 81-station nominal B-rep transfer sweep over 200 mm has no internal feeder-solid intersection; that check did **not** include the first-article track and carrier. A later cross-assembly audit found a **34,050 mm³** transfer-bridge/guide overlap and a **74,910 mm³** mid-transfer payload/guide overlap. An unselected integration revision clears exact-solid overlap in four STEP states and 81 sampled transfer stations by lifting a guide window, shortening stop proxies and removing duplicate study solids. R1 leaves a **10 mm unsupported handoff gap**. A separately saved R2 bridge revision meets the support lip in sampled nominal states, with four STEP states and 81 sampled transfer stations again showing no exact-solid penetration. The bridge-extension/guide-lift sequence, continuous swept volume, contact load, tolerance and power-off restraint remain unverified. Under *assumed*, not manufacturing, perturbations, the earlier feeder-only interval screen leaves 3.5 mm side and 2.5 mm end-stop clearance. The state/fault analysis finds a decisive unresolved condition: a payload partway between cassette and cradle lacks a demonstrated positive restraint after power loss. Neither the pusher actuator nor launch locks, contact loads, jam recovery or a working four-cell 1U/3U feed mechanism is selected. The integration audit records the boundary of the new check. A 6U cell is a later expansion. These follow-on files do not change the Gen5 result or establish P-POD-like reliability.

The subsequent four-cell R0 study positions two 1U and two 3U inert envelopes in staggered and paired fixed cells. Each unselected layout has 13 editable FreeCAD and STEP states with zero represented nominal solid penetration; four lateral transfer paths and a stowed-carrier docking path were sampled separately. In the staggered layout the final cell has only 330 mm of modeled acceleration path before the brake boundary, requiring 5.56 g *ideal mean* acceleration for 6 m/s. Paired sides give every cell at least 900 mm, reducing that largest ideal mean value to 2.04 g, without changing the 1,800 mm rail. Both layouts span 560 mm laterally before host mounting or access. These are geometry and kinematic screens, not proof of achievable speed, safe contact, power-off retention, carrier docking accuracy or host compatibility.

The provisional seven-concept weighted screen applies the selected 35/25/20/10/10 weights for fault safety, fit/mass, positioning, cycle/power and manufacture. A captive lead screw ranks first as an **analysis reference** at 3.650/5; varying each unmeasured ordinal criterion by one point gives 2.650–4.650. That interval overlaps the spring-follower and rack-pawl alternatives. None has passed the mechanism-level four-cell, power-off-restraint, load or installed-resource gates, so the result does not select a feed mechanism. Alternative concepts remain in the laboratory archive for comparison.

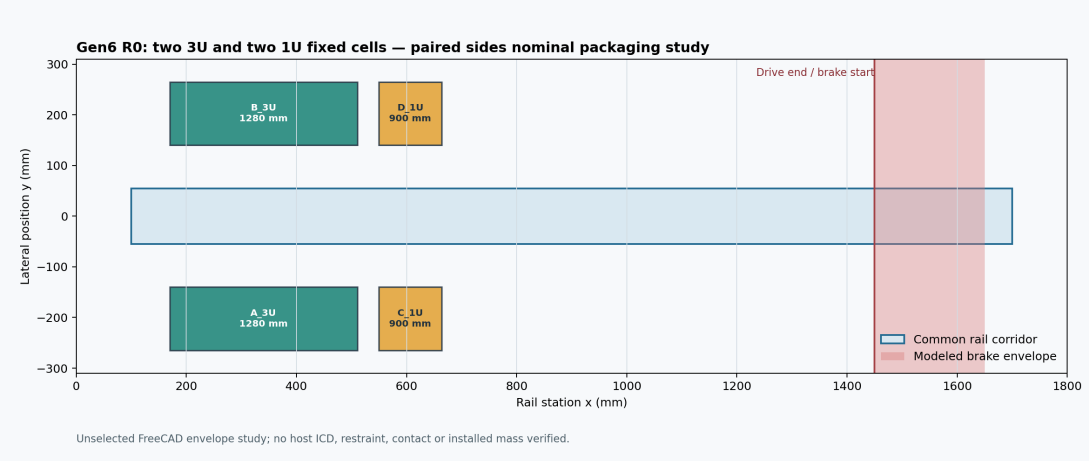


Figure 19: FreeCAD-derived paired four-cell plan. Payload cells are envelope proxies; motion, loaded clearances and host fit are unverified.

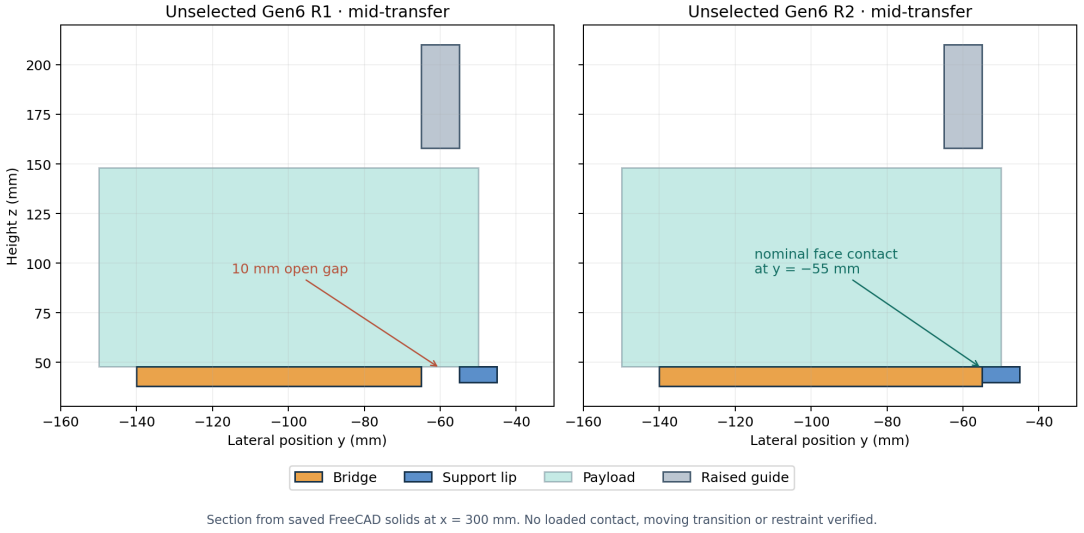


Figure 20: FreeCAD-derived handoff section. R1 has a 10 mm open support gap; R2 has nominal face contact. Bridge loads and transition motion are not verified.

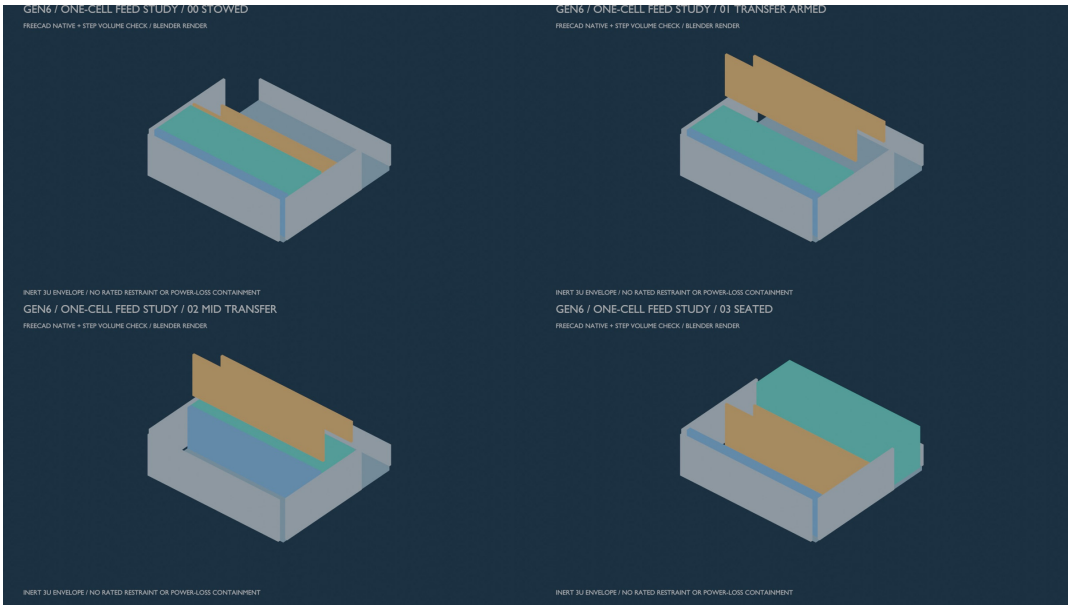


Figure 21: STEP-checked FreeCAD geometry rendered in Blender for four Gen6 one-cell transfer states. Moving hardware and fault containment are incomplete.

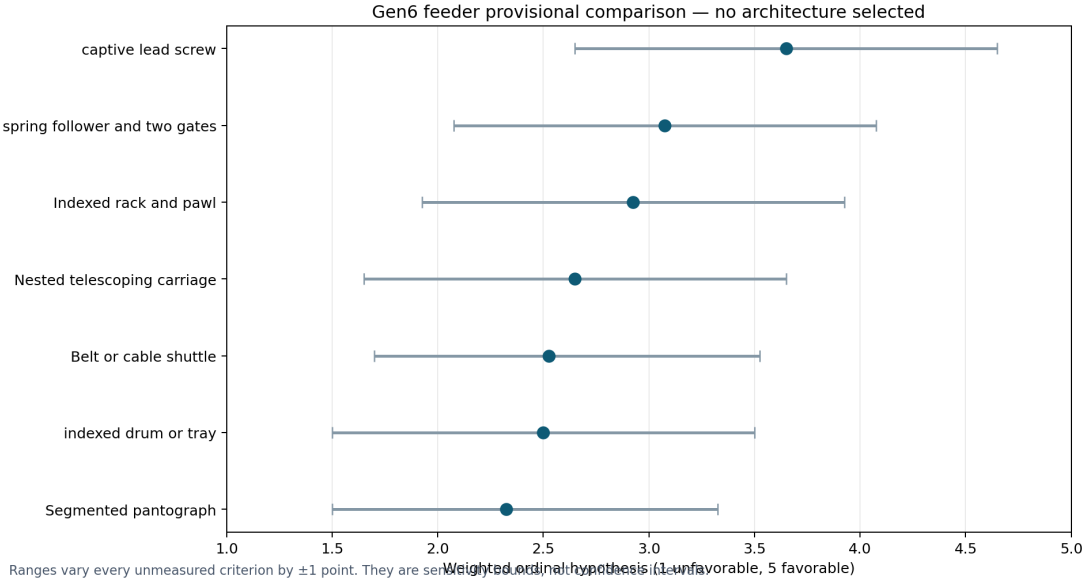


Figure 22: Provisional feeder ordinal scores and one-point sensitivity intervals; not measured reliability or mass.

These screens make the next research questions specific, while preserving Gen5 as the sole evaluated thesis configuration. They do **not** close Gen6's mission value, full installed mass, moving-contact/release design, electrical and thermal ratings, qualification or build authorization. The claim and gate disposition traces the open items for examination.

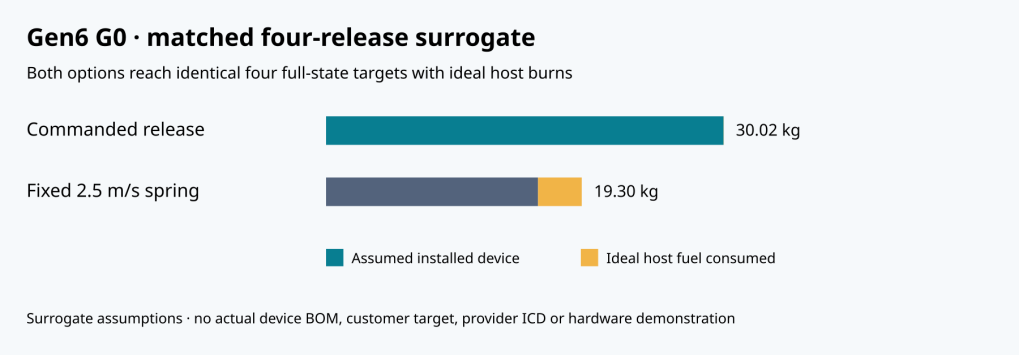


Figure 23: Surrogate four-release comparison. Both strategies reach the same ideal mathematical targets; installed mass assumptions decide this narrow comparison.

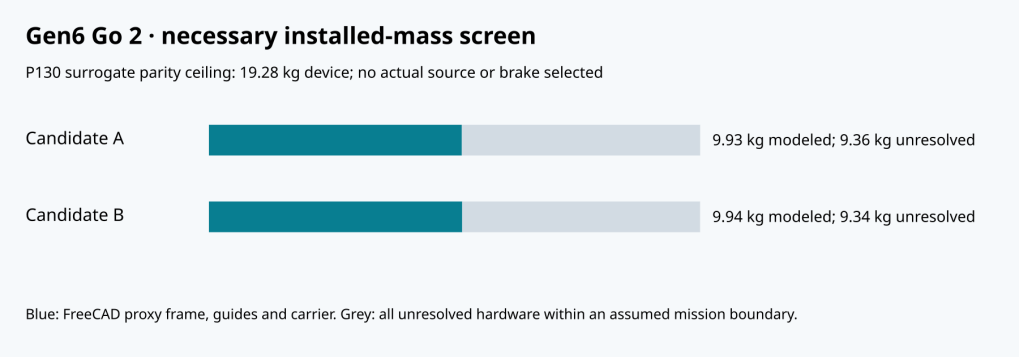


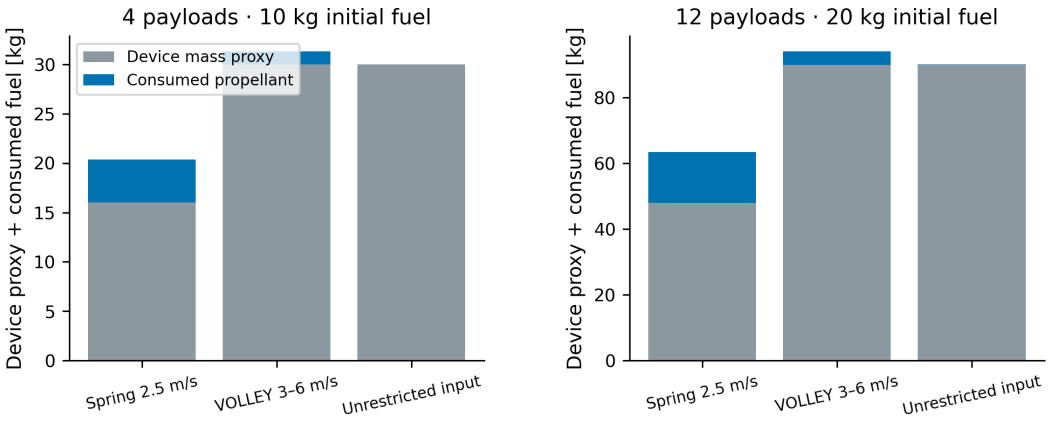
Figure 24: Gen6 first-article partial CAD material mass versus the remaining surrogate allowance. Unmodelled installed items are excluded.

C.2 Coupled four/twelve-payload construction

The local coupled-delivery supplement adds host pre-burn, finite-mass shot recoil, 20 s coast, transfer/recovery impulses, propellant accounting and final host-state restoration. It uses a 500 km, 97.4° surrogate and constructed 3–9 m/s prograde target increments, with common state assessment at 90000 s. A 3–6 m/s imposed input can reach the targets with host assistance; no motor capability or delivered-state accuracy is demonstrated.

The twelve-payload spring case exceeds a 10 kg reserve but completes with 20 kg initially available, consuming approximately **15.35 kg**. The corresponding 3–6 m/s case consumes approximately **4.05 kg**. The assumed device proxies are 48 and 90 kg, so the fuel saving does not recover the mass penalty. No optimized customer mission or commercial advantage

is established. Conditional adaptive interval bounds for nominal point centers exclude the first 20 s near-field phase and assume a 1 cm numerical state-error bound; they are not a physical clearance or collision-probability result.



Constructed common-epoch states, ideal host impulses and fixed-axis J2. Device proxies are 16/30 kg per 4 payloads, not a designed installed BOM. Loaded reserves differ between panels; no customer service or commercial advantage proven.

Figure 25: Declared mass proxies and consumed fuel; no installed-system advantage inferred.

C.3 Seven-concept engineering evidence

The local feeder review includes all seven candidates, 1U/3U native CAD, STEP states, material-only BOM/CG/inertia, 108 analytical load cases, 12 tolerance stacks and 36 conditional state/fault records. Six native CalculiX runs examine an assumed clamped storage-floor fixture; they do not analyze actual joints or qualify the assembly. The spring sizing fails higher resistance cases and the adverse alignment stack loses clearance. A 972-case rectangular capture-tooth sensitivity reaches approximately 1.97 GPa nominal root bending under the largest calculated mean arrest force, exceeding even the assumed 1000 MPa yield band. Actual material allowable, peak impact force and positive engagement are unknown; this is a redesign finding.

The connected fork-frame correction is distinct from proven mechanical load transfer. Gate/latch engagement, front-barrier handoff, continuous return/indexing, purchased components, actual host mounts and loaded restraint/contact remain open. Ordinal safety-led trade intervals overlap; no concept meets the evidence gate for final baseline selection. The report and source/CAD archive preserve this bounded result without claiming a completed or manufactured feeder.

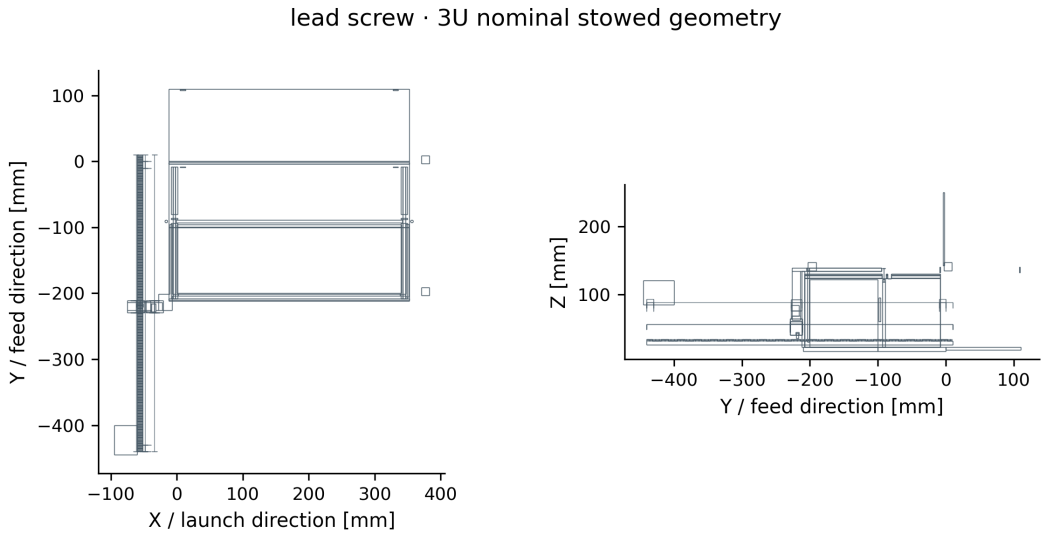


Figure 26: Independent STEP-derived nominal lead-screw reference geometry; unselected mechanism with open handoff/contact verification.

C.4 Continuous prescribed transfer and servicing disposition

The separately versioned CT-01 local report provides 22 native/STEP states and continuous interval checks for prescribed rectilinear 1U/3U transfer, receiver capture, fork withdrawal and reset. The storage gate now retracts downward through a floor slot; the 120 mm stroke requires under-deck accommodation. Receiver locks use separate motion bands. All represented nominal primitive pairs clear over the 20 phase cases, with 60 independent native B-rep poses. This does not validate actuator joints, loaded fit, contact or actual receiver-to-sled custody.

A separate enlarged 16 × 6 mm rack-root coupon has one modeled face-contact pose and six native CalculiX fixture cases. Its peak stress remains unconverged and its actual material/impact/attachment are unqualified. The 972-case compliant-stop investigation retains 226 cases exceeding a proposed 24 mm total allowance; 148 of 243 local fit sensitivities are nonpositive. False receiver-lock indication, uncontrolled gate closure and missing external sled custody remain explicit physical counterexamples. That dated coupon study supplied no feeder selection or hardware release; the later FM08-R3 design decision is reported separately.

The limited replenishment concept defines a closed cooperative cartridge, provisional datums, capture/retention and contamination interfaces, operations and recovery requirements. Its 108 capture and 36 insertion sensitivities retain 18 failed fits. Because the

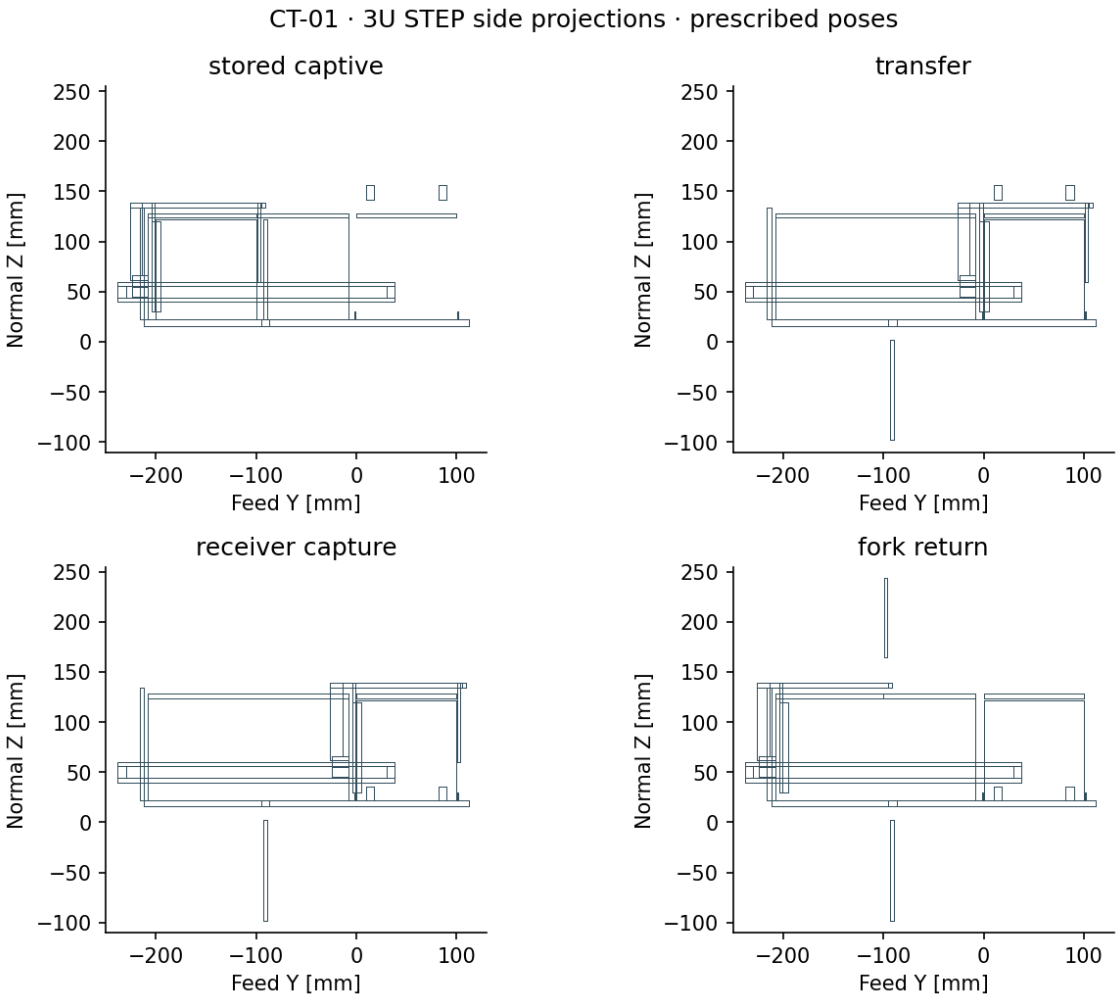


Figure 27: CT-01 STEP side projections with common axes; nominal prescribed motion only

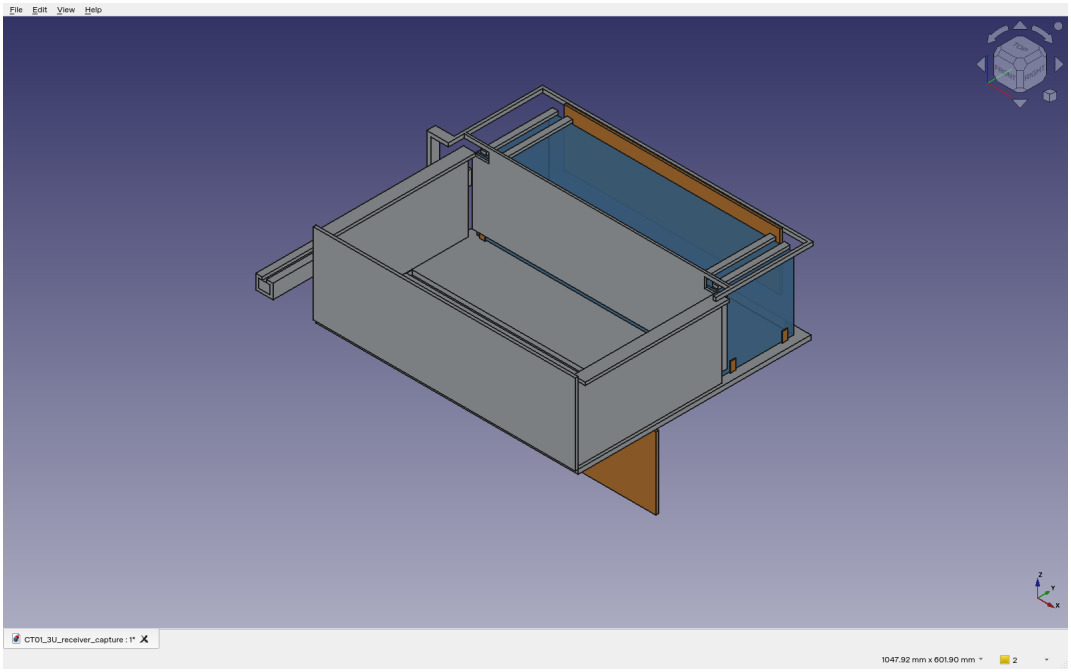


Figure 28: Genuine Python-hosted FreeCAD Qt/Coin viewport of the CT-01 receiver-capture pose. Source document and capture provenance are retained; this is a native application view, not hardware or a completed mechanism.

baseline lacks removable-cartridge integration, real servicing access and verified double custody, operational replenishment is a no-go for this configuration. Ground loading remains the assumed campaign boundary. The limited study is complete as a bounded negative feasibility/interface finding; it is not a docking or robotic hardware demonstration.

C.5 Complete native event replay and precision budgets

The separately versioned CNR-01 native campaign report checks the complete prescribed four/twelve-payload host-burn, finite-mass release and recovery sequence from a single initial host state using Orekit 13.1.6. Fifty-one native cases include five nominal baselines, two step refinements and 44 deterministic command-error probes. Native rocket mass loss, recoil and payload propagation agree with the original source at nominal payload endpoints to within 8 µm; the twelve-case step refinement changes payload endpoints by less than 17 µm. These are software consistency results under a fixed-axis J2 model, not physical delivery accuracy or a rigorous continuous error enclosure. Both implementations use DOP853; commanded controls and constructed targets are shared inputs.

A common +0.1% release-speed bias produces approximately 1.53 km position error in the imposed VOLLEY cases. A first-order allocation of the entire illustrative 100 m / 0.1 m/s full-state budget to common speed bias alone is approximately 0.0059%, with all other

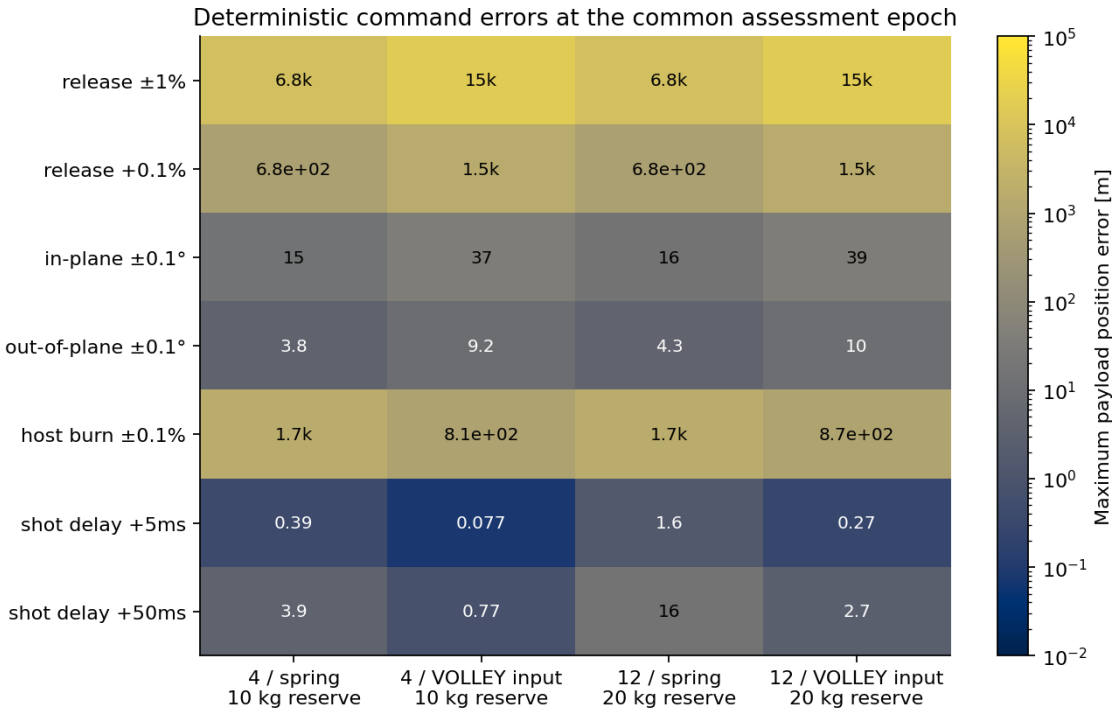


Figure 29: Common-epoch position sensitivity to stated deterministic probes; surrogate missions and no navigation feedback

errors zero. This is not a customer requirement or supported actuator precision. Three hundred twenty-four constant-speed photogate interval cases and 108 dynamic metrology cases expose calibration, timing, thermal, mean-to-exit-speed and frame-compensation dependencies. A longer optical baseline can worsen the exit-speed bound when relative acceleration persists. Post-release measurements cannot correct an already released passive payload.

The twenty equivalent-thrust time screens apply only to the frozen impulse policy. At 5 N, the twelve-spring control plan requires approximately 6135 s of thrust-on time before a 3600 s final release, while the imposed 3–6 m/s case requires approximately 1488 s. These necessary totals omit finite-burn targeting, slew, coast, plume and operational scheduling. Other host-assisted spring strategies may improve this result; no universal service advantage follows. Device mass proxies continue to exceed the fuel saving. Actual installed hardware, finite-burn optimization, navigation feedback and full near-field integration remain unresolved.

C.6 Finite-duration host targeting

The separately versioned FBR-01 finite-force report targets actual finite-duration resultant-force windows with integrated mass consumption. Eighteen solutions vary the two-window

fraction, numerical step and fuel reserve; 46 native Orekit runs include nominal replays, refinements and deterministic force/release/initial-state probes. Preparation from −900 s, 20 s post-release guards and recovery to 4200 s are counted. All payloads retain the common 90000 s assessment epoch. The inputs remain imposed release velocities and a fixed-axis J2 point-mass model.

At the 35%/35% policy, four-release spring/VOLLEY inputs require 13.36/6.50 N peak and 6.056/1.895 kg fuel. Twelve-release inputs require 33.83/18.00 N and 23.122/6.123 kg. The twelve-spring 20 kg reserve is exhausted at 3509.06 s; subsequent formal trajectory is diagnostic only. With a common 30 kg reserve and 100 N limit, both twelve-release controls reach the constructed full-state targets. The 17.460 kg propellant saving remains smaller than the 42 kg device-proxy penalty.

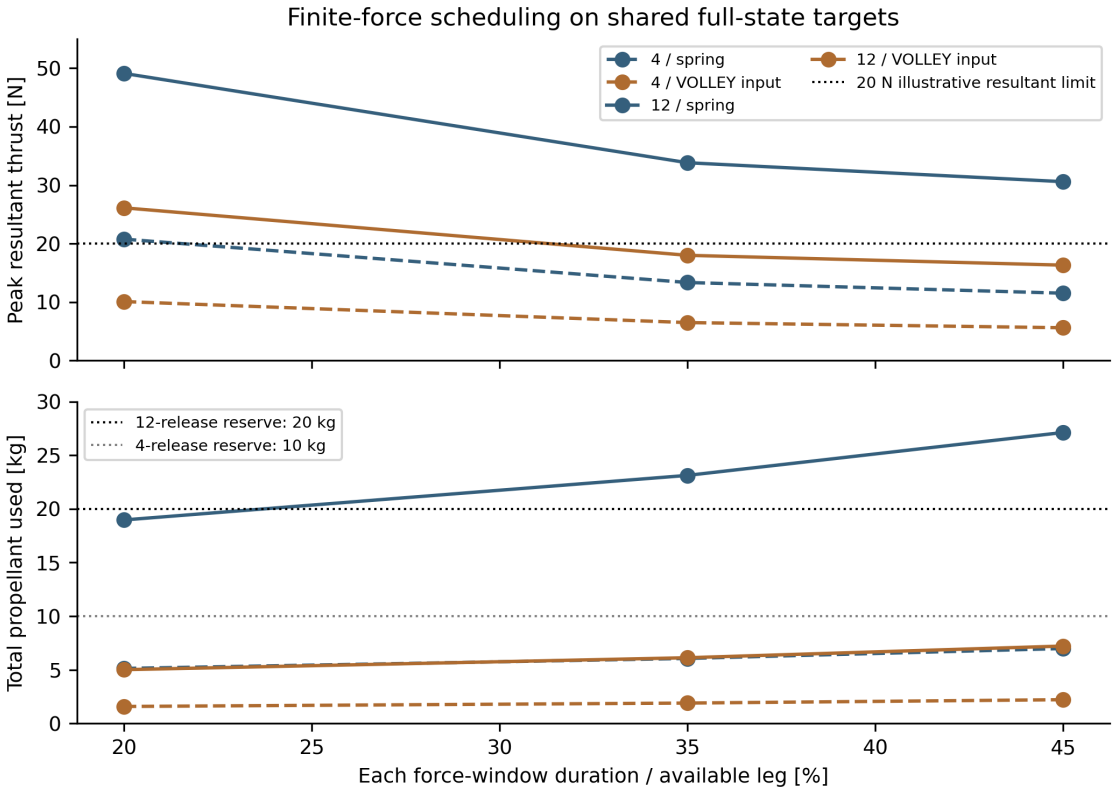


Figure 30: Retargeted finite-force schedules under common mission targets. Each point is a separate solution; limits and hardware resources are surrogate assumptions.

Native nominal terminal differences stay below 23 µm and eight numerical refinement differences below 13 µm. These verify numerical consistency, not real delivery accuracy. A single fixed-thruster interpretation requires nearly 9°/s mean direction reversal in some 20 s gaps, with no settling allowance. Opposed thrust allocation might avoid whole-host reversal, but its plumbing, mass, plume geometry and attitude requirements are undefined.

Host compatibility, physical near-field safety, navigation feedback, installed hardware and the motor/circuit rating remain open; no operational service is established.

C.7 Feeder actuation and revised custody geometry

The separate FM-02 engineering review replaces the tall receiving latch with a 12 mm underfloor entry frame, two positive endpoint crosspins and a physically represented sled. Twenty-eight native FreeCAD/STEP poses and 26 continuous prescribed rigid-translation phases have no represented positive-volume penetration. The storage crossbolt and each receiving pin obstruct the tested adverse motions. Axial pin capture, actual release actuators, loaded contacts and complete deployment/reset remain unimplemented; no hardware containment or manufacturing release follows.

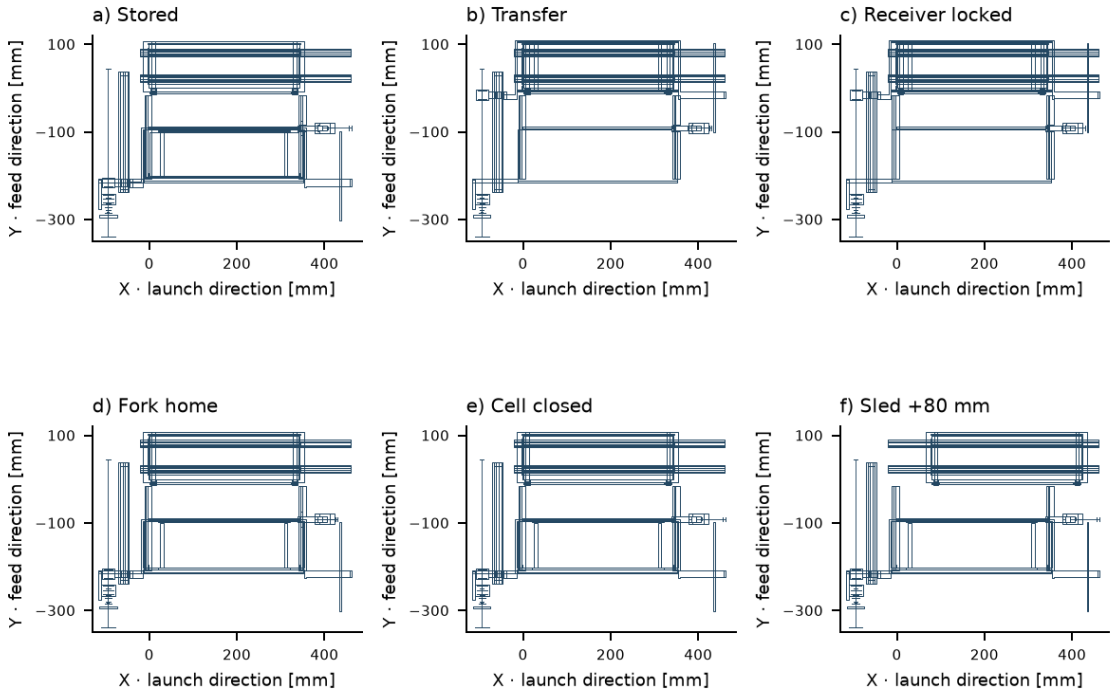


Figure 31: FM-02 captive handling sequence from actual STEP exchanges. The payload remains retained during the illustrated 80 mm sled translation and return; no launch is represented.

The integrated ball-screw candidate is a separate ambient component screen and is not installed in this CAD. A quintic rest-to-rest law, moving load, drag, rotational inertia and component limits are checked in 972 scenarios. Polynomial extrema and exact piecewise integrals reproduce every decision; at 801 samples the maximum relative torque and input-energy-proxy discrepancies are 8.58×10^{-8} and 1.08×10^{-6}. These are numerical model errors, not physical uncertainties. The sampled 10, 15 and 20 s cases meet the declared ambient screen; 5 s cases fail.

The study retains 115 of 324 nonpositive fit residuals, 12 of 54 winding-temperature failures, 96 no-stop brake sensitivities and 72 finite runouts above the proposed 24 mm allowance. Actual component temperature domains differ, and the two-node thermal model is uncorrelated. A slow 103.0 s handling assumption does not fit eleven later twelve-release coasts of 92.182 s; a faster 74.5 s assumption fits, without actuator, attitude, indexing or full campaign validation. These findings define remaining engineering requirements. At that dated stage, all seven feeder alternatives remained unselected; the subsequent FM08-R3 selection is reported separately. Gen5's computational result is unchanged.

C.8 Feeder geometry, component stiffness and contact limitations: 21 October 2026

The separate FM06-R1 record introduces integral support tines, a 12 mm front-restraint stroke and longer receiver guides. Thirty native FreeCAD/STEP poses, seventy individual part exchanges and 28 continuous prescribed rigid-translation checks retain no represented nominal positive-volume interference. An edge-only backstop/rail junction, missed by solid-validity checks, is corrected and all seventy component contact surfaces pass a closed, consistently wound mesh audit. These are reference geometries, not manufacturing drawings or physical qualification.

The separate FM07-R5 release/reset geometry adds captive receiver pins, a hinged exit gate and empty-carrier reset. Forty-six nominal cycle states and 34 additional rotary samples have no represented collision above the stated numerical threshold. Forty continuous linear sweeps check the prescribed translations; the rotary phases remain discretely sampled. Twenty native/STEP assemblies and 74 part exchanges pass independent readback. The fork/tines and receiver entry-bar material are unchanged from FM06; its isolated fixture results remain distinct from the changed assembled body and hinge. Single-cell mass proxies are 3.7624 kg for 1U and 6.1566 kg for 3U, excluding payload and unmodeled installed hardware. Neither this geometry capture nor the FM06 contact study closes the all-seven architecture-selection gates.

Twelve native FLR03 cases analyze the actual FM06-R1 receiver solid. Forty historical FLR01 cases provide a separately identified revision comparison. Under 1,000 N reference loads and assumed steel stiffness, the 3U one-dog mean loaded-face displacement changes from 1.8829 mm for FM02 to 0.05998 mm for current FM06. The historical FM03 result is 0.05811 mm and is not substituted for the current revision. The largest final FLR03 mesh-to-mesh compliance change is 0.200%. Both pin bores are fully clamped, so the reduction

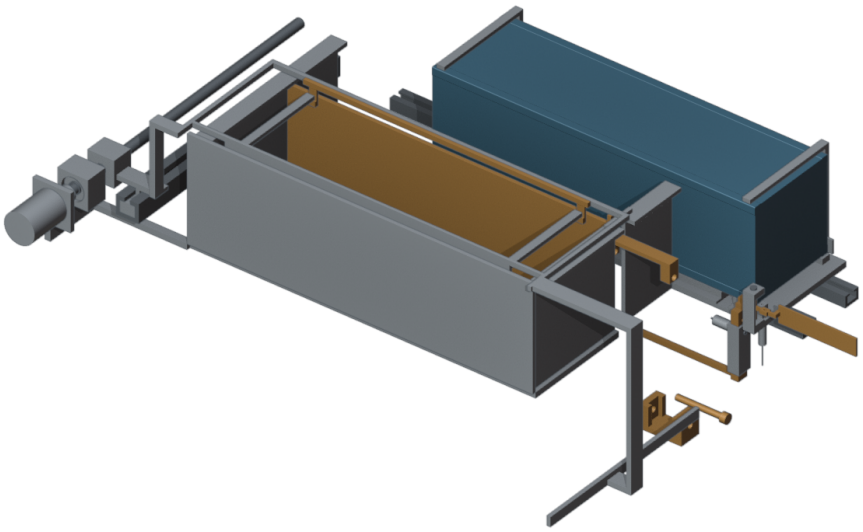


Figure 32: FM07-R5 exit-open state rendered from native FreeCAD geometry in Blender Workbench. Prescribed geometry only: no launch dynamics, loaded-fit or manufacturing acceptance.

applies to this component fixture and does not bound assembled stiffness. Reaction force and moment balance are checked; no selected alloy allowable or assembled pin-contact acceptance follows.

Four native unit-load tine meshes yield 0.0101446 mm/N mean end compliance at the finest 53,650-element mesh, with a final 0.0334% change. Forty linear-scaled load sensitivities retain large-deflection and gap failures. At 6 kg and 1 g, an assumed equal load split gives approximately 0.298 mm displacement per tine against a nominal 0.6 mm slot gap; one-tine loading nearly consumes that gap before root/frame flexibility and tolerances. The clamped isolated tine is not a qualification model.

Longer guides reduce the bounded rigid rocking excursion, but the tighter fit creates a separate manufacturing problem: 1,117 of 1,215 deterministic adverse width/alignment/thermal/deflection combinations have nonpositive residual width. The count is not a statistical hardware failure rate.

The FCE12 contact-verification study uses nine disjoint convex native payload cells and exact orthogonal mechanism cells, with full original body mass and inertia. Internal kg/mm/s scaling is checked against a no-contact SI replay. Fourteen analytical planar-gap controls detect imposed 10 µm–1 mm overlaps; the largest checked reported-gap error is

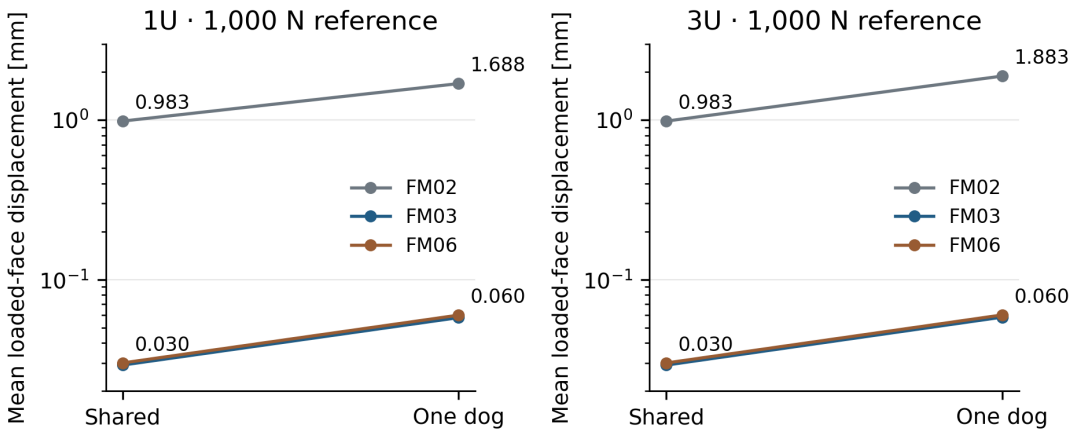


Figure 33: Native component-fixture compliance comparison. The logarithmic axis covers historical and current receiver revisions under identical reference loads; the supports are idealized clamped bores.

4.28 nm. This is a calibration of the stated planar controls, not a general contact-accuracy claim.

Ten of twelve native sequences reach the 26.5 s transfer/capture/withdrawal endpoint; deliberate jam and stuck-pin cases abort at 11 s. Independent full-solid checks cover 84 targeted poses. Two residual material intersections, at most 0.000124 mm³, admit a nonintersecting pose after a 1 µm Y translation; original trajectories and residuals remain preserved. These discrete checks do not certify continuous loaded clearance.

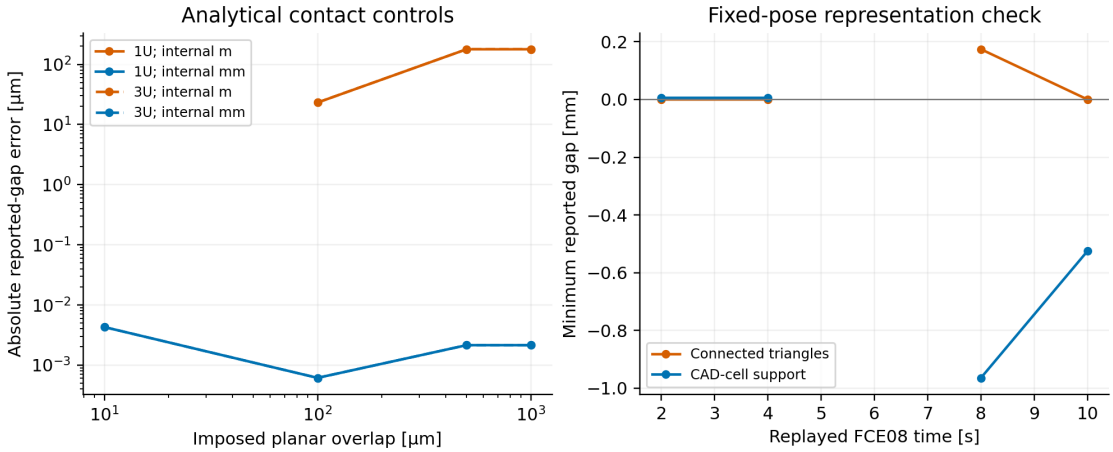


Figure 34: Analytical planar-gap controls and fixed-pose replay in the native contact solver. The millimetre-coordinate calculation resolves the imposed overlaps; calibration does not establish final-state convergence or measured precision.

The 250-to-125 µs final centre-of-mass differences are 1.509 mm for 1U and 1.466 mm for 3U; the provisional 0.05 mm numerical criterion is not met. Impulsive rigid-contact peaks remain step-sensitive and are not design loads. The capture begins with the storage gate open and does not include free separation or a complete reset. Source-rated installed actuation,

actual loaded joints, payload interface approval and mixed-size installed integration remain engineering acceptance requirements. The evidence-gated seven-architecture downselect remains open.

C.9 End effects and research evidence

The targeted reference review checks selected methods/results/conclusion passages in the Japanese original of the Halbach end-effect study (DOI 10.1541/ieejias.143.570; English translation DOI 10.1002/eej.23450). Its 2D/JMAG comparison and conditional coil-coverage cancellation method provide context, not finite-width 3D or hardware validation of VOLLEY. No numerical correction is imported. Full text of the close double-magnetic-circuit CubeSat actuator article (DOI 10.1109/TIE.2021.3134060) was not available in the bounded lawful-access search; it remains metadata/abstract-only and supplies no quantitative performance or interface claim.

C.10 Continuous rotary-clearance check: FM07-R6

The RVC01 native CAD study examines the four rotary phases separately from the R5 sampled-angle evidence. Near the closed state, the R5 gate lower edge and sled deck both lie at 22 mm. Native distance is zero while their intersection volume is zero: this is surface contact, not a demonstrated positive-volume collision.

FM07-R6 is a separate candidate revision. Raising only the gate panel lower edge by 0.25 mm, while retaining its upper edge, hinge and locks, removes 41 mm³ of steel per gate (0.32185 g at the assumed density). Adaptive angle-interval checks use native B-rep distances, a rigid-rotation displacement bound and a 0.00001 mm numerical guard. All four complete 0–90° opening/closing screens pass; the smallest guarded conservative bound is 0.0098846 mm. This is a numerical nominal-geometry bound, not a manufactured clearance or certified kernel error enclosure.

Eight full assembly exports and two individual gate STEP files pass separate CadQuery/OpenCascade readback. Eight source/delta checks confirm that the other components are unchanged and that the corrected gate is a material subset of R5. The separate importer uses the same kernel family. The change does not establish loaded clearance, tolerance acceptance, actuation, contact convergence or architecture selection.

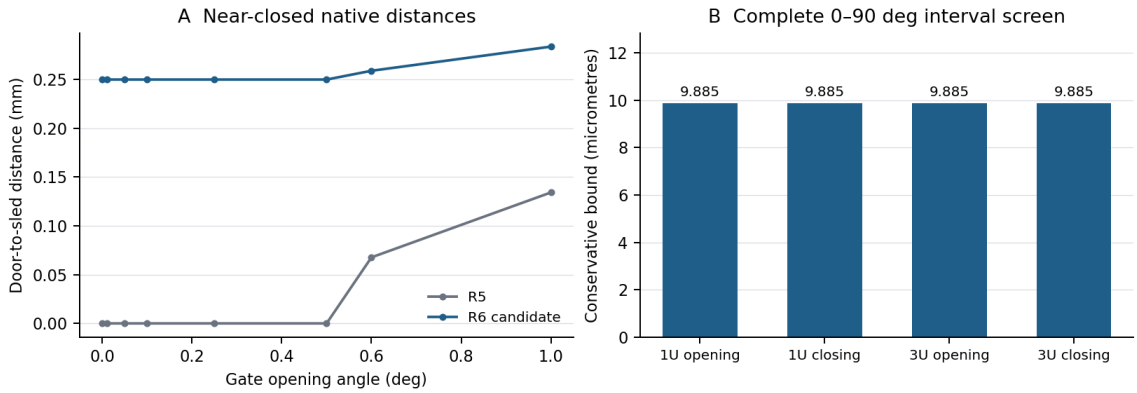


Figure 35: Native gate/deck distance and conservative complete-angle screen. The 0.25 mm panel relief differs from the smallest global guarded bound. Source: RVC01, FM07-R6; nominal prescribed geometry.

C.11 Contact-model numerical diagnosis: FCE13

The FCE13 diagnosis retains FM06-R1 geometry and physical assumptions. Six additional native Chrono runs reach the common 26.5 s endpoint. Instrumentation reproduces the original 3U 250 µs trajectory byte for byte. Solver-budget use concentrates during feed deceleration, before receiver capture. Neither warm starting nor a preconditioned Barzilai–Borwein solver establishes final-state convergence. Replacing the rectangular core mesh with the exactly equivalent native box also fails the pose screens; at 125 µs, the representation alone changes the endpoint by 0.755420 mm and 0.603748°.

Table 13: FCE13 common-endpoint 250-to-125 µs comparisons. Acceptance limits remain 0.05 mm COM and 0.05° orientation.

Refinement comparison	COM change [mm]	Orientation change [deg]	Retained pose screen
Original PSOR, 1U	1.508584	0.639678	Fail
Original PSOR, 3U	1.466380	0.232303	Fail
Native BB, 3U	1.926456	0.120098	Fail
Equivalent box core / PSOR, 3U	0.716162	0.564333	Fail

Work differences remain below the separate 1% screen, which does not establish pose or contact-load convergence. Two terminated APGD pilots are archived and excluded from endpoint comparisons. Native solver residuals have different definitions across PSOR and BB and are not directly interchangeable. These observations quantify tested numerical sensitivity; they are not bounds on the exact solution. No numerical correction is adopted,

no rigid-contact force rating is established, and FM06 contact results are not transferred to FM07-R6.

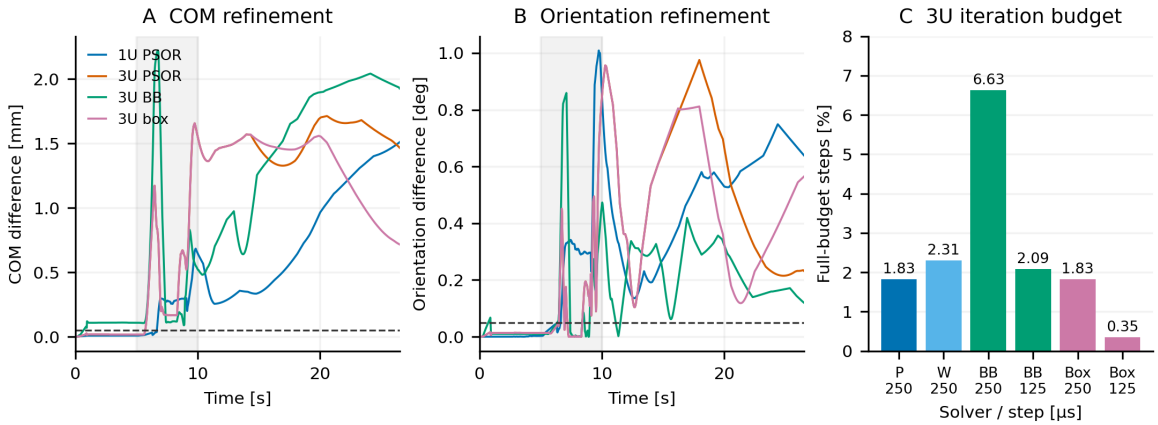


Figure 36: Common-time refinement differences and full 2,000-iteration-budget use in native Chrono. P: PSOR; W: warm-start PSOR; BB: preconditioned BB; Box: equivalent box core with PSOR. Shading marks feed deceleration. Dashed lines retain the 0.05 mm and 0.05° screens. A full-budget solve may first meet tolerance at its final iteration; budget counts alone are not residual convergence tests. Source: FCE13, FM06-R1.

C.12 Lunar and upper-speed compatibility bounds

The standalone analytical study separates carrier propulsion and payload release. An ideal coplanar, tangential patched-conic screen gives Earth departure 3.106427 km/s, lunar arrival excess speed 0.830117 km/s and direct capture into a 100 km circular lunar orbit 821.239408 m/s. It does not include finite sphere-of-influence or epoch targeting. Capture into a 100 × 10,000 km ellipse initially needs 306.643674 m/s, but later circularization at the same perilune restores the same total impulse. No net saving follows when the endpoint is held fixed.

Ideal engine-only escape thresholds from the three staging examples are 676.620, 523.149 and 162.024 m/s. The assumed 200 m/s allowance cannot establish escape from the first two. Reference-sphere grazing is a geometric limit, not a targeted disposal solution; no-shot release-recoil credit is zero. Useful lunar lifetime, navigation covariance, a complete carrier campaign, resource/thermal closure and controlled disposal remain unresolved.

For a 4 kg payload and 300 kg retained host, 1 km/s relative release needs at least 1.973684 MJ ideal added relative energy and 2.012598 km constant-force stroke at a 25 g study ceiling. This rejects the declared compact/low-acceleration combination; the acceleration is not a payload rating. Source, moving-pusher, friction, charging and catcher burdens remain

additional. The near-term moderate-speed Gen6 article and upper-speed investigation are therefore separate studies.

Nine algebraic checks reproduce 182 numerical fields. The source package retains a historical C0-S2 criteria-hash mismatch explicitly; it does not certify that earlier criteria provenance or a perturbed lunar mission. The new calculations are conditional analytical bounds, not a selected provider, hardware rating or flight validation.

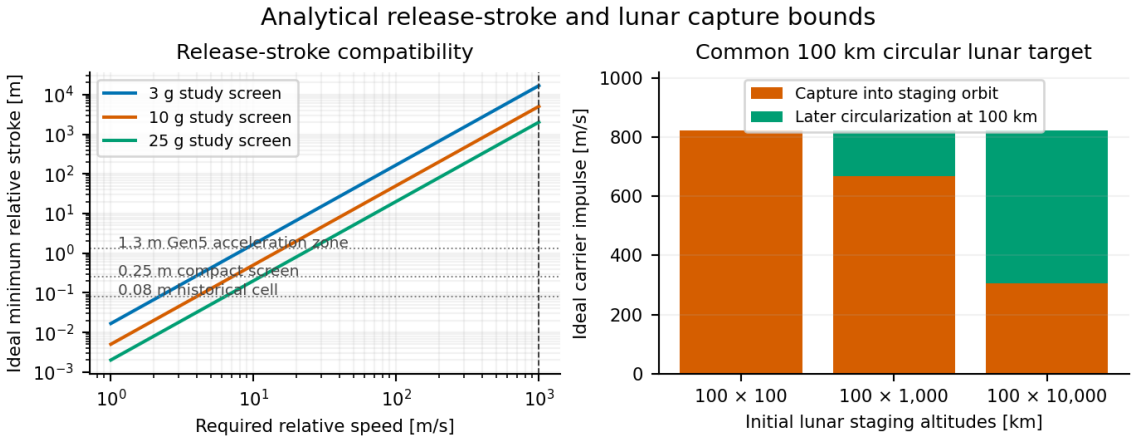


Figure 37: Ideal stroke compatibility and capture/circularization accounting for a common final lunar orbit. Assumed payload/host masses and acceleration screens are stated in the source package; no useful lifetime or provider performance is established.

C.13 Seven-feeder common-boundary CAD comparison: FDR09, 22 October 2026

The FDR09 development study replaces unequal one-cell/four-cell mass comparisons with a declared four-payload inventory: two 1U and two 3U satellites, one common receiver and separate local feed stations. Fixed-cell candidates model a 900 mm shared receiver travel on 1,600 mm rails; the drum retains a distinct index/extraction arrangement. Common 3U outer handling packages use passive 1U inserts, each with modeled mass 0.185933 kg. This packaging is not an optimized storage-density result or a qualified spacecraft interface.

Table 14: FDR09-R1 partial mass accounting. Catalog options are not integrated or qualified. Neither column establishes full installed mass, CG or inertia for FDR09-R1; missing hardware is not assigned zero mass.

Candidate	Represented CAD material [kg]	Listed catalog options [kg]	Mandatory eligibility
Captive lead screw	18.606	4.230	Not eligible

Candidate	Represented CAD material [kg]	Listed catalog options [kg]	Mandatory eligibility
Spring / dual gate	18.915	6.002	Not eligible
Rack / pawl	18.618	5.830	Not eligible
Indexed drum	14.821	1.932	Not eligible
Pantograph	18.060	6.002	Not eligible
Nested telescope	18.132	4.230	Not eligible
Belt shuttle	17.915	5.830	Not eligible

The source-backed option screen contains 384 time/station evaluations using manufacturer motor/gear/controller data. Listed cycle times remain conditional actuator options, not verified complete feed cycles. Spring force deficiency, a 35.5 mm pantograph output gap, telescopic component intersections and common-path interferences remain in the source records. Under the original FDR09 eligibility rule, mandatory gates preceded weighted ranking. None of its seven R1 candidates was eligible on 22 October, so that study supplied no controlled selection. The subsequent FM08-R3 design decision uses the explicitly stated academic-design boundary below.

The separate lead R2 correction trims obstructing guides/capture racks, shortens the screw blank, shifts its end bearing and relieves a guide-bearing housing. The checked nominal rectangular payload sweep changes from eight intersections to zero; the identified 16.176779 mm^3 housing interference becomes zero. The corrected custom-material subtotal is 18.492547 kg. A 0.25 mm radial screw-bearing gap remains; positive bearing support, full-carriage clearance, loaded/tolerance/fault behavior, insert lock/return custody and full installed accounting are not accepted. R2 is separate from the seven R1 comparisons and from FM07-R6.

Ten native/STEP pairs cover seven R1 candidates, corrected lead R2 and two insert coupons. A separate CadQuery/OpenCascade import checks all ten exchanges. For the eight full comparison assemblies, imported solid counts and volumes agree with the native records under the declared 1 ppm relative serialization-volume criterion. This is not a manufacturing tolerance or a different-kernel certification. The original spring serialization diagnostic and its sampled-surface evidence remain recorded.

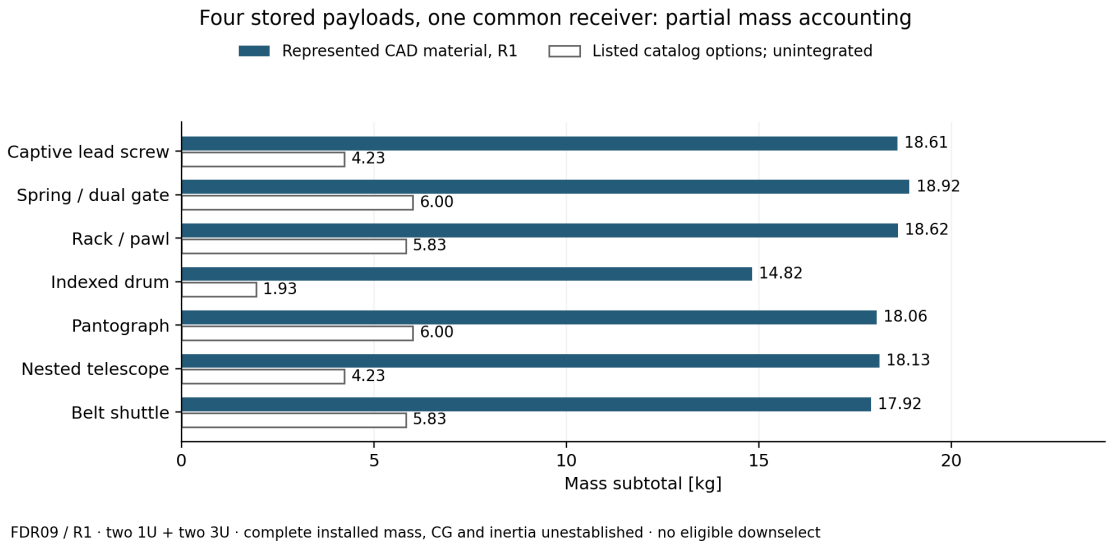


Figure 38: CAD material subtotals and separately listed catalog hardware options for the same four-payload boundary. Neither is a full installed-system mass; all candidates retain unpassed mandatory gates. Source: FDR09-R1.

C.14 Selected feeder design: FM08-R3, 23 October 2026

FM08-R3 selects independently driven, guided lead-screw cassettes feeding one indexed reusable receiver. The reference inventory is two passive 1U and two passive 3U envelopes. Fixed 1U spacers remain in the cassettes; retractable receiver stops distinguish the aft datums while retaining a common leading datum. No loose insert is released with the spacecraft. Rack-and-pawl transfer is retained as the runner-up.

The selected-design record contains editable native CAD, STEP exchanges, nominal operating-envelope checks, the declared tolerance and analytical load screens, supplier-reference actuator sizing and executable sequence/fault checks. The selection defines a computational academic mechanism; it is separate from a qualified manufactured mechanism.

C.14.1 Represented geometry and mass properties

The stowed native assembly contains 154 named parts. Its represented custom material is **25.364310 kg**, with material CG **[592.142, 51.908, 54.706] mm** in the assembly X/Y/Z frame. X is the launch/index direction, Y is the local feed direction and Z is the assembly vertical. The bounding envelope is **1,610 × 877.8 × 388 mm**, or **548.344 L** of bounding-box volume. This is an accommodation requirement, not an approved host deck allocation.

The represented material inertia about that CG is:

$$I_{\mathrm{CG,represented}} = \begin{bmatrix} 1.180194 & -0.520026 & 0.013393 \\ -0.520026 & 4.180394 & -0.028242 \\ 0.013393 & -0.028242 & 5.022837 \end{bmatrix} \ \mathrm{kg\,m^2}.$$

The five complete assembly states each contain 154 physical features and 154 STEP solids. Inclusive exact-solid checks find no positive-volume interference. The installed operating-envelope matrix passes all 115 sampled cases, with 18 additional local payload-motion samples. Separate CadQuery/OCP imports pass the five STEP validity, solid-count and volume checks, with maximum relative volume difference 2.94×10^{-14}. FreeCAD and CadQuery use the OpenCASCADE kernel family, so this is a separate import/serialization check rather than independent-kernel certification. Nominal sampled clearances do not establish continuous tolerance, loaded-contact or manufactured acceptance.

These properties exclude payload internals and catalog hardware masses. The separately listed primary-drive catalog subtotal is 4.473 kg, giving a **29.837310 kg partial feeder accounting boundary** before omitted equipment. Harness, fasteners, gate/retainer actuation, the launch drive, power source and final thermal hardware are not silently assigned zero mass. Full installed mass, CG and inertia require their resolved installation; the later mission cases therefore use explicit omitted-equipment allowances.

C.14.2 Architecture trade and sensitivity

The trade assigns 35% to safety/fault control, 25% to packaging/mass, 20% to positioning/custody, 10% to cycle/power and 10% to manufacturability. Scores are structured engineering judgments from the documented mechanisms and available evidence, rather than measured reliability or success probabilities.

Table 15: FDR10 architecture selection under the declared safety-led weights. The scores are judgments, not hardware measurements.

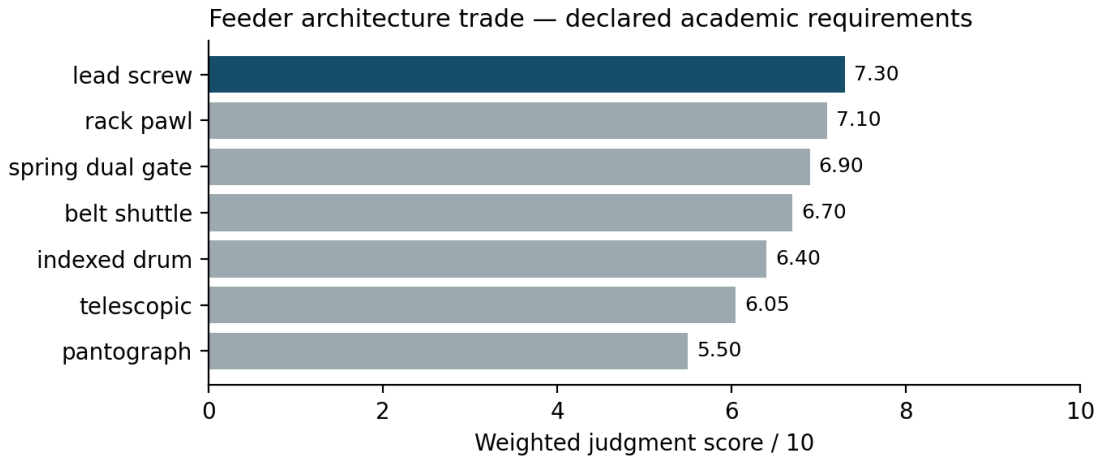
Architecture	Weighted judgment / 10	Design disposition
Guided lead screw	7.30	Selected for controlled pusher displacement and channel containment
Rack and pawl	7.10	Runner-up; engagement, backlash and tooth-load detail remain relevant

Architecture	Weighted judgment / 10	Design disposition
Spring and two gates	6.90	Compact alternative; force, governing, reset and gate synchronization need redesign
Belt shuttle	6.70	Alternative; tension, compliance and retention require attention
Indexed drum	6.40	Alternative; common indexing and extraction increase integration demands
Nested telescope	6.05	Alternative; extension overlap and alignment require further design
Pantograph	5.50	Alternative; joint count, near-flat loads and output continuity penalize this case

The 0.20-point lead over rack and pawl is modest. Equal weights give a three-way tie at 7.20; packaging-emphasis weights favor the spring/two-gate concept, and cycle-emphasis weights favor rack/pawl. The decision is consequently a requirement-dependent engineering compromise. The rejected concepts retain their research value and source history; the selection does not establish a universally optimal mechanism.

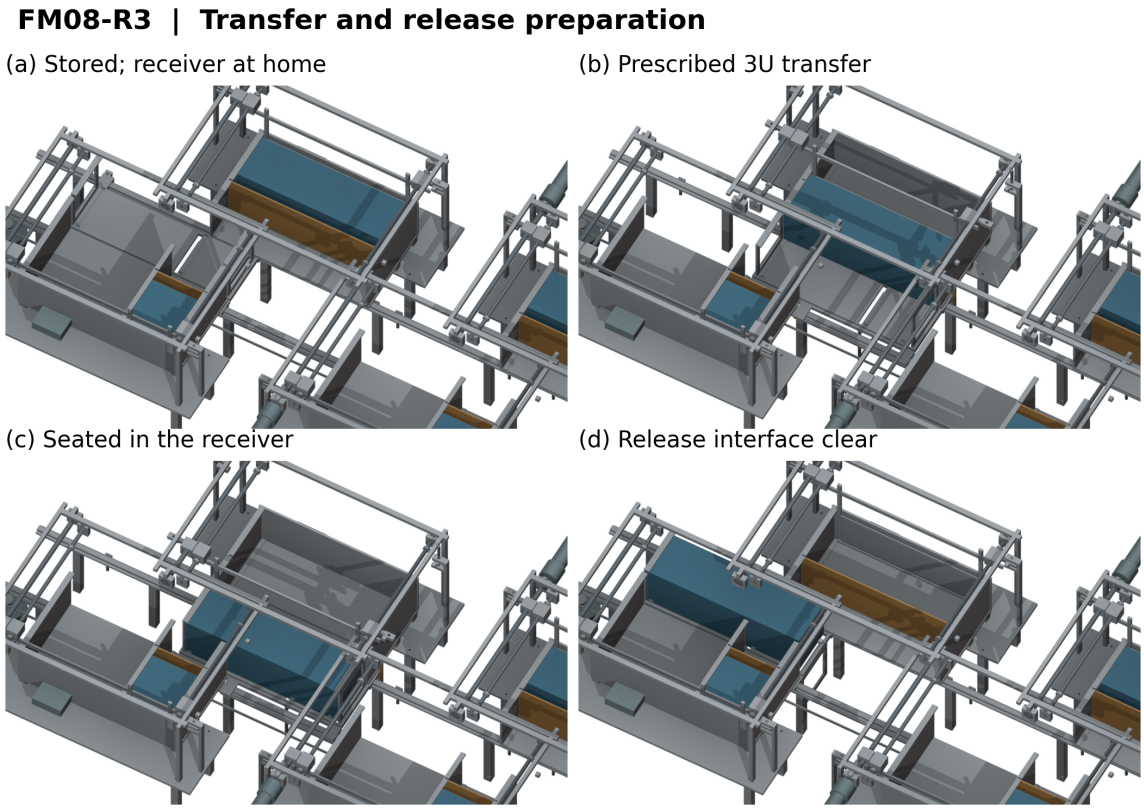
C.14.3 Operating sequence and fault containment

The nominal sequence is: retain the payload in its cell; index and dock an empty receiver; establish docking, empty-receiver and closed-far-stop proof; open the cell boundary; advance the guided pusher through the capped channel; establish seated-presence and settling proof; close the near-side receiver capture tabs; withdraw the pusher; close the cell; then permit launch preparation. The receiver toothed-belt index drive uses a split clutch: independent disengagement proof is required before launch arming. After launch, independent stopped-carrier proof precedes clutch engagement and engagement proof, then empty return and home/reset. The next feed is inhibited until this sequence completes.



Safety 35% · Packaging 25% · Positioning 20% · Cycle/power 10% · Manufacture 10%
Engineering judgments; scores do not measure reliability. FDR10 / D27-R2.

Figure 39: FDR10 weighted architecture judgments. The seven values follow the stated weights; sensitivity changes the preferred architecture. This plot contains no experimental reliability data.



Cropped native CAD views; neighbouring structures may be truncated.

Prescribed states, without dynamic or time fidelity. The pusher commands transfer; channel stops and required proofs define the delivery datum.

Figure 40: FM08-R3 prescribed storage, transfer, seated and launch-preparation states. CAD-derived visualization; fixed 1U spacers remain in storage. The image does not represent measured motion or a physical test.

A power loss, jam, timeout, contradictory sensor state or double-feed request inhibits release and commands a retained safe state. A failed cell may be isolated only after a verified clear common path and supervised recovery. A bridge obstruction inhibits the entire shared path. The capped channel, closed receiver far stop, guides, hard stops and a normally engaged static brake support the containment concept. The pusher does not pull the spacecraft: low-friction forward motion inside the channel and end-stop impact remain unmeasured. A jammed payload requires supervised recovery; pusher reversal alone does not demonstrate payload retrieval. Screw self-locking is not assumed.

The deterministic state model passes **32,821 enumerated logic cases**, including the four-cell sequence, return/home/reset, fault injection, forbidden commands, repeated release and isolated-cell continuation. The twelve-cell extension passes a separate 98,325-case logical campaign. Nominal payload transfer requires a coast condition, with host burn off and a stable-host proof; launch also requires 5 s with the host burn off. An unexpected burn inhibits the cycle and resets the guard. These are tests of the implemented logical contracts. Actual latch engagement, truthful independent sensor evidence and brake response remain physical interface contracts; the test count is not a hardware reliability estimate.

C.14.4 Conditional actuator and structural sizing

The local drive reference is a FAULHABER 3242G024BX4 motor with a 32GPT 9:1 gearhead and MC5005S controller. The receiver uses the same motor family with a nominal 59:1 gearhead, actual ratio 59.4. The captured supplier data and parameter classifications [13–17] are retained in the local package. Supplier ambient ratings do not establish vacuum, launch or thermal qualification.

The local prescribed transfer is 200 mm in 30 s; the four-cell receiver indexes up to 900 mm in 20 s. The calculation assumes 50 N nominal resistance, a 100 N adverse screen and 0.50 gear efficiency. The trapezoidal-thread model includes flank friction, bearing drag and reflected rotating inertia; the receiver model includes tangential belt force at the declared 12 mm pulley radius and reflected inertia. Belt tooth engagement, pretension and backlash are not supplier-qualified by this sizing model. Motor continuous torque is 0.0418 N·m in the captured data.

Table 16: Conditional drive sizing at the declared motion profiles. The 100 N receiver result uses approximately 97% of the catalog continuous torque; the small margin depends on the assumed resistance and efficiency.

Motion and resistance	Peak motor torque / N·m	Peak motor speed / rpm	Peak modeled current / A
Local feed, 50 N	0.01485	3600	0.448
Local feed, 100 N	0.02627	3600	0.724
Receiver, 50 N	0.02038	4254	0.581
Receiver, 100 N	0.04058	4254	1.069

A pinned-pinned Euler calculation for the 7.4 mm screw root and a 350 mm supported span gives 2372 N critical load and 0.00407 mm calculated axial shortening at the 100 N adverse condition. The restraint-pin screen uses declared material allowables and 5/10/25 g sensitivity loads; these are study bands rather than provider qualification. The corridor calculation retains **0.623 mm** after its stated manufacturing/alignment/thermal and 75 N guide-deflection stack. It applies to the modeled corridor direction, rather than qualifying every bearing fit, end barrier or spacecraft gauge.

The local 0.12 N·m static-brake reference exceeds the ideal frictionless 155 N backdrive demand by a factor of 2.43. The receiver separately uses a 2 N·m static-brake reference against a 1.2 N·m holding demand at 100 N and 12 mm pulley radius, giving a static ratio of 1.67. These comparisons do not establish dynamic arrest distance, impact load, engagement time or wear. Source friction, gear efficiency, actual assembled stiffness and environmental behavior require correlation before a hardware performance rating.

C.14.5 Evidence disposition

The selected feeder is an academic design baseline with controlled geometry and bounded verification. The earlier FM06-R1 rigid-contact model remains numerically unconverged; its endpoint differences and impulsive peaks are not used as release-precision, latch-load or contact-force acceptance for FM08-R3. Prescribed kinematics and analytical load assumptions are explicitly distinguished from a calibrated dynamic contact model. The Gen5 motor speed and installed-mass results do not transfer to this new geometry.

C.15 Twelve-cell geometric scaling and current mass boundary

The detailed twelve-cell extension uses six 1U and six 3U envelopes, repeated from the four-cell module with a 4,200 mm rail and 3,500 mm receiver travel. Its actual stowed native assembly contains **398 named parts** and **72.730463 kg represented custom material**. The corresponding material CG is **[1883.728, 52.088, 57.072] mm**. Its bounding envelope is **4,210 × 877.8 × 388 mm**, or **1,433.869 L**; these dimensions are a host-accommodation requirement, without provider acceptance.

Both twelve-cell native/STEP states contain 398 physical features and 398 solids. Their inclusive exact-solid checks report zero positive-volume interference; all twelve modeled rectangular payload swept corridors also clear the represented solids. These checks cover nominal represented geometry, rather than tolerance distributions, dynamic contact or provider accommodation.

The selected primary-drive catalog subtotal is 10.601 kg, giving **83.331463 kg partial feeder accounting**. Neither this subtotal nor the four-cell 29.837310 kg subtotal is full installed mass. The matched mission cases add declared omitted-equipment allowances of 10, 25 or 50 kg, covering still-unspecified installation burden rather than assigning it zero mass. The resulting four-cell device assumptions are 39.837, 54.837 and 79.837 kg; the twelve-cell assumptions are 93.331, 108.331 and 133.331 kg. They are sensitivity cases, not weighed or supplier-complete assemblies.

The 3,500 mm receiver travel retains the prescribed 90 mm/s peak speed by using 77.778 s per full index. The conservative feeder-only allocation is 223.556 s per payload, or 2,682.667 s for twelve. This allocation excludes the launch, braking, charging and attitude-control requirements; the native orbital follow-up checks the declared schedule separately.

C.16 Matched mission retarget with the selected feeder

FBR-02 retargets the common full position/velocity delivery states and common final epoch for both host-assisted spring separation and imposed 3–6 m/s VOLLEY release. The 300 kg base-host, 220 s specific impulse, J2 force model, 100 N maneuver-force ceiling and 10/30 kg initial fuel for four/twelve spacecraft are declared study inputs, rather than provider

ratings. Every release has a 5 s zero-force preparation guard and a 20 s post-release guard. No achieved electromagnetic release speed is inferred.

All ten selected-mass targeting cases meet their declared force, fuel and state-residual criteria. The 26 native Orekit runs and ten campaign conservation/guard checks pass. The maximum nominal terminal position discrepancy is 1.973×10^{5} m and the maximum velocity discrepancy is 2.184×10^{8} m/s. Maximum conservation residuals are 7.96×10^{13} kg for mass and 6.61×10^{1} kg·m/s for momentum. These measure numerical replay consistency, rather than physical delivery precision. Reducing maximum step from 60 to 30 s changes terminal position by at most 4.72 μm for four spacecraft and 8.26 μm for twelve. Separate frozen-command 3U mass probes from 4 to 6 kg miss the target by 19.543 km and 43.018 km, showing the need for accurate payload mass and retargeting. The following table uses the central 25 kg omitted-equipment allowance for VOLLEY. The spring reference is also evaluated at longer maneuver windows because it does not inherit the shared feeder handling time.

Table 17: FBR-02 conditional matched-state mission resources. The parameter f sets the fraction of each transfer leg assigned to each of its two maneuver windows; it is not an overall mission duty cycle. Device masses are explicit partial-feeder-plus-allowance assumptions, rather than supplier-complete installations.

Manifest and release option	Assumed device mass / kg	Host peak force / N	Propellant used / kg
Four, spring, f = 0.35	16.000	13.356	6.076
Four, VOLLEY + 25 kg allowance, f = 0.35	54.837	6.979	2.039
Twelve, spring, f = 0.10	48.000	93.657	17.676
Twelve, spring, f = 0.20	48.000	52.044	19.769
Twelve, spring, f = 0.35	48.000	35.858	24.100
Twelve, VOLLEY + 25 kg allowance, f = 0.10	108.331	51.884	4.852

The four-cell minimum pre-release coast is 268.5 s, leaving 160.5 s after its 108 s feeder-only allocation. The twelve-cell coast is 241.818 s, leaving 18.263 s after 223.556 s of feeder handling. Charging, launch, braking, reset and attitude-control operations must also fit their actual available intervals; a positive feeder allocation residual alone does not close the complete operational schedule.

At the central allowance, VOLLEY uses 4.036 kg less propellant for four spacecraft and 12.824 kg less for twelve than the lowest-consumption tested spring policy. Its assumed device dry-mass penalties are 38.837 kg and 60.331 kg, respectively. The dry burden therefore exceeds these propellant-use reductions. The spring f = 0.35 policy also needs less peak force than the twelve-cell VOLLEY policy. The result establishes a conditional propulsion-resource trade, without an installed-mass, optimized wet-mass, cost or customer-service advantage.

C.17 Resource requirements and lunar navigation sensitivity

The separate SYS-01 and LNS-01 package extends the requirements analysis without assigning an achieved launch rating. SYS-01 evaluates 108 energy/charging cases, 18 constant-force stroke cases, 36 single-axis host-impulse cases and 54 optimistic heat-rejection cases. Retained host mass is 300 kg. Net efficiency of 25/50/75% and charging power of 25/50/100 W are declared investigation bands.

For a 4 kg payload at an imposed 6 m/s relative release, ideal reduced-mass energy is 71.053 J per shot. Four independent identical 4 kg shots at assumed 25% net efficiency require at least 1.137 kJ from the bus and 22.74 s of charging at 50 W, before additional source/catcher burdens. This homogeneous sensitivity case is distinct from the mixed 1U/3U reference manifest. Recoil work and energy are accounted for; a centered point impulse produces zero single-axis angular impulse, while an offset changes the angular demand. Flexible-host response, real attitude control and thermal correlation are outside that point-impulse model.

LNS-01 evaluates lunar osculating-periapsis sensitivity at three staging apoapsides and three assumed uncertainty bands. The model uses two-body motion about a spherical lunar reference and a point release at apolune. Each case has 4,096- and 8,192-sample checks, with finite-difference refinement of the local state Jacobian. The uncertainty inputs are

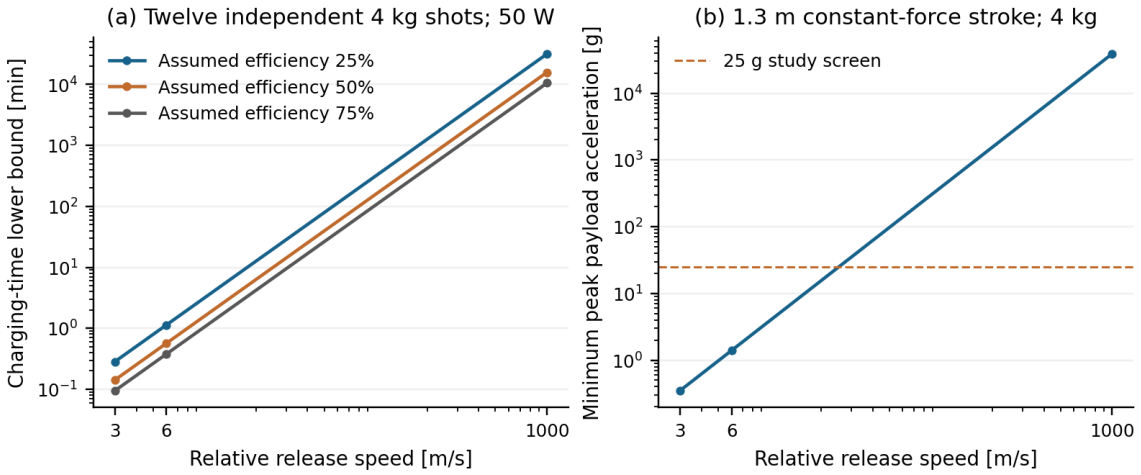


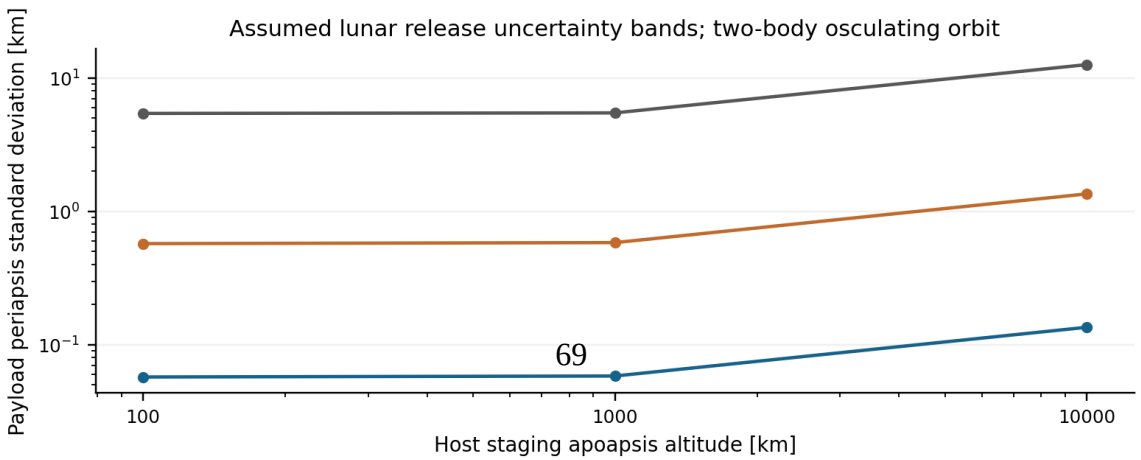
Figure 41: SYS-01 charging and acceleration requirements. Left: twelve independent 4 kg shots, 50 W charging and assumed net efficiencies. Right: the constant-force lower bound for a 1.3 m stroke. The 25 g line is a study ceiling, not payload qualification; source, catcher and environmental burdens remain additional.

declared independent zero-mean Gaussians, rather than measured navigation or actuator statistics.

Table 18: LNS-01 assumed independent uncertainty bands. The distributions are study choices, rather than flight specifications.

Assumed band	Position σ	Velocity σ	Pointing σ	Relative-speed σ
Tight	10 m	0.01 m/s	0.01°	0.1%
Middle	100 m	0.1 m/s	0.1°	1%
Wide	1 km	1 m/s	1°	5%

Under the middle band, the linearized periapsis standard deviation is approximately 570 m, 581 m and 1,357 m for staging apoapsides of 100, 1,000 and 10,000 km. The corresponding 8,192-sample standard deviations are 573 m, 583 m and 1,353 m. The local Jacobian refinement checks pass the stated 10^{-4} relative criterion. These comparisons support the sensitivity calculation within its model; lunar terrain, nonspherical gravity, external perturbations, coverage, communications and useful orbital lifetime are excluded.



C.17.1 Native lunar coast and gravity-model sensitivity

The separate LNC-01 native GMAT R2026a study propagates three prescribed lunar orbits from 23 October 2026, using DE421 Earth/Sun point perturbations and lunar gravity degree/order 20, 40 and bounded 80 comparisons. All 18 prescribed attempts have captured native histories. Of twelve cases requesting 180 days, seven terminate at the 1,737.4 km spherical-radius condition and five reach 180 days; the six higher-degree/alternative-field cases complete their requested 30 days. The stated 90° inclination is relative to MJ2000Eq, rather than a lunar-polar mission plane.

For the 100 km circular 90° reference, the spherical stop occurs at 115.859 days with degree/order 20 and 117.984 days with 40. At the common 90-day checkpoint, reducing maximum integration step from 120 to 60 s changes position by 0.0575 m, whereas degree 20 to 40 changes it by 10.35 km. Degree 40 to 80 differences at 30 days remain 619–1,398 m across the three orbit cases. Small integration-step sensitivity therefore does not establish gravity-model convergence. These finite-horizon coasts exclude topography, stationkeeping and delivery targeting; neither a spherical stop nor reaching 180 days establishes useful mission lifetime or continuous terrain clearance.

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